Hybrid MoM-PO Solver for Antenna Placement Analysis

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Abstract

English

This project focuses on designing a combined MoM and PO solver that focuses on identifying the antenna position in complex electromagnetic fields. The aim is to accurately calculate scattered magnetic fields when an incident field interacts with various structures, combining the high accuracy of MoM for fine details and the computational efficiency of PO for larger structures. The solver is implemented using MATLAB and validated against FEKO simulation results to ensure accuracy.

afrikaans

Hierdie projek fokus op die ontwerp van 'n gekombineerde metode van oomblikke (MoM) en fisiese optika (PO) oplosser wat fokus op die identifisering van die antennaposisie in komplekse elektromagnetiese velde. Die doel is om verspreide magnetiese velde akkuraat te bereken wanneer 'n invallende veld met verskeie strukture in wisselwerking tree, wat die hoë akkuraatheid van MoM vir fyn besonderhede en die berekeningsdoeltreffendheid van PO vir groter strukture kombineer. Die oplosser word geïmplementeer met behulp van MATLAB en bekragtig teen FEKO-simulasieresultate om akkuraatheid te verseker.

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Nomenclature

Variables and functions

λ	Wavelength
$\mathbf{B}$	Magnetic flux
$\mathbf{H}$	Magnetic field Intensity
$\mathbf{E}$	Electric field Intensity
μ	Permeability
ξ	Permittivity or small number
$\mathbf{D}$	Electric flux
∇	Derivative
$\mathbf{M}$	Magnetic current density
$\mathbf{J}$	Electric current density
$\mathbf{Z}$	impedance Matrix
$\mathbf{k}$	Wave vector
ω	Angular frequency
p_v	Electric charge density
$\hat{\mathbf{n}}$	Normal unit vector
$\mathbf{G}$	Green's Function
$\mathbf{V}$	Excitation vector
$\mathbf{r}$	Observation point
$\mathbf{r}'$	Triangle Centroid/ source point

List of Abbreviations

Acronyms and abbreviations

PO	Physical Optics
MoM	Method of Moments
FEKO	Field Simulator for Electromagnetic Field Computation
MATLAB	Matrix Laboratory
PEC	Perfect Electric Conductor
TEM	Transverse Electromagnetic
PDEs	Partial Differential Equations
CEM	Computational ELectromagnetics
EM	Electromagnetics
TE	Transverse Electric
GO	Geometric Optics
RWG	Rao - Willton - Glisson

Chapter 1

Introduction

The goal of this project is to develop a scattering solver that calculates the scattered magnetic field by employing Physical Optics (PO) and the Method of Moments (MoM). The aim is to calculate the radiated magnetic field when an antenna is located in proximity to a larger structure. This involves formulating the problem in terms of integral equations, where the surface currents created on the object by the incoming field are determined using Hybrid MoM-PO. The MoM is used for fine details of an antenna and PO is used for the electrically large structure it is placed upon. These surface currents are then utilized to obtain the scattered magnetic field. The scope includes the mathematical formulation of the problem, the implementation of numerical methods for solving the integral equations, and the validation of the solver against known solutions or experimental data. The envisioned approach combines theoretical modeling, computational techniques, and validation to create a tool for predicting scattered magnetic fields in antenna placement applications.

1.1. Background

Antenna placement analysis plays a vital role in modern communication systems, radar technologies, and electromagnetic compatibility studies, as it directly impacts the performance and efficiency of these systems. A key aspect of this analysis is the accurate calculation of scattered electromagnetic fields, which is essential for optimizing antenna design and performance, particularly in complex environments where interactions with surrounding structures significantly affect field distribution. Traditional methods for this purpose include the MoM and PO [3].

Full-wave methods (like MoM) approximate Maxwell's equations numerically and are highly accurate in modeling fine details of electromagnetic interactions [2], but it becomes computationally expensive when applied to electrically large structures due to its dense matrix formulations and high computational resource requirements as a result Asymptotic methods (like PO) uses simpler estimates for faster results on the electrically large structures and provides a more efficient solution by approximating the scattered fields,

but it becomes inaccurate for electrically small structures. To address these limitations, a Hybrid MoM-PO solver is developed (it has already been proposed by others), combining the strengths of both methods.

Hybrid methods, such as the MoM-PO approach, partition the problem domain into regions best suited for each technique, MoM is employed for areas requiring detailed full-wave solutions, such as the antenna's fine structures, while PO is used for electrically large structures [4], smooth regions where an asymptotic approach is sufficient. The goal of this project is to develop a scattering solver that accurately calculate the scattered magnetic field when a known incident magnetic field interacts with an object. The formulation involves setting up integral equations to determine the surface currents created on the object by the incoming field using the Hybrid MoM-PO method.

Once these surface currents are computed, they are used to determine the scattered magnetic fields. The scope of this work includes not only the mathematical formulation of the integral equations but also the implementation of numerical methods to solve them and the validation of the solver against known solutions or experimental data. This approach integrates theoretical modeling, computational techniques, and practical validation [5], aiming to create a tool for predicting scattered magnetic fields in various practical applications, thereby enhancing the accuracy and efficiency of antenna system design and placement in complex environments.

1.2. Problem Statement

The main problem addressed in this project is the development of a scattering solver capable of accurately calculating the scattered magnetic field when a known incident magnetic field interacts with an object. The challenge is to combine the MoM for the antenna's fine details and PO for the electrically large structure upon which it is placed, ensuring computational efficiency and accuracy.

1.3. Objectives

To solve the problems mentioned above sec 1.3, the following goals must be met:

1. To formulate the scattering problem using integral equations.
2. To implement the integral equations, PO solver for plane wave scattering and MoM in MATLAB.
3. To implement a Hybrid MoM-PO solver in MATLAB for solving the integral equations.

4. To validate the solver against known solutions or experimental data using both MATLAB and FEKO simulations.

5. To calculate and validate far-field and near-field patterns using MATLAB by comparing them with FEKO simulations to ensure accuracy.

6. To integrate shadowing effects into the Hybrid MoM-PO code for accurate modeling of real-world electromagnetic scenarios.

1.4. Scope

This report covers the theoretical formulation of the scattering problem, the formulation of the numerical solver, and the numerical implementation of the Hybrid MoM-PO solver in MATLAB, the validation process using FEKO, and the analysis of results obtained from simulations and experiments. The study focuses on evaluating the performance and accuracy of the Hybrid solver for various antenna placement scenarios. However, it does not account for the effect of shadowing in the calculation of scattered electromagnetic fields, which could impact the results in environments where shadowing plays a significant role. Future work may consider incorporating shadowing effects to provide a more comprehensive analysis of scattered fields in complex scenarios. The report also covers the calculation of both far-field and near-field.

1.5. Report Overview

- Chapter 1 introduces the project, covering the background, problem statement, objectives, and scope. It explains the purpose of the study.

- Chapter 2 reviews the literature on the Hybrid MoM-PO solver, placing the study within the context of existing research.

- Chapter 3 describes the Physical Optics (PO) Solver, explaining how it calculates scattered fields for large structures and how it is implemented in MATLAB.

- Chapter 4 explains the Method of Moments (MoM) Solver, focusing on how it models surface currents on detailed antenna structures.

- Chapter 5 discusses how the Hybrid MoM-PO solver presents results for antenna placement analysis.

- Chapter 6 summarizes the findings, discusses their impact, and suggests future directions for the work.

Chapter 2

Literature Review

2.1. Maxwell's equations

Electromagnetics involves the study of the interactions and effects of electrically charged particles, both at rest and in motion. These interactions are fundamentally described by Maxwell's equations, presented in equations 2.1 to 2.4, which will be utilized in the calculation of electromagnetic scattering. Along with these equations, the constitutive relations $\mathbf{B} = \mu\mathbf{H}$ and $\mathbf{D} = \varepsilon\mathbf{E}$ are also considered.

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} - \mathbf{M} \tag{2.1}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \tag{2.2}$$

$$\nabla \cdot \mathbf{D} = \rho_v \tag{2.3}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{2.4}$$

2.2. Physical Optics (PO)

PO provides an approximation method for calculating the current induced on electrically large surfaces, such as a parabolic reflector 2.1. In this approach, currents are considered only on the illuminated region of the surface, while shadowed regions are assumed to have no current, as shown in Figure 2.2. This current density, $\mathbf{J}_{\mathrm{PO}}$, can be defined by:

$$\mathbf{J}_{\mathrm{PO}} = \begin{cases} \hat{\mathbf{n}} \times \mathbf{H}_{\mathrm{total}} & \text{in the illuminated region} \\ 0 & \text{in the shadowed region} \end{cases} \tag{2.5}$$

$$\mathbf{J}_{\mathrm{PO}} = 2(\hat{\mathbf{n}} \times \mathbf{H}^i) \tag{2.6}$$

where $\mathbf{H}_i$ is the strength of the incoming magnetic field at the surface S, and $\hat{\mathbf{n}}$ is the unit vector that is perpendicular to the surface. To determine the radiation characteristics

(such as Magnetic field, near-field, and far-field patterns) of a reflector, we must know the current density $\mathbf{J}_{\mathrm{PO}}$ induced on its surface. Using the expression

$$\mathbf{J}_{\mathrm{s}} = \hat{\mathbf{n}} \times (\mathbf{H}^i + \mathbf{H}^r), \tag{2.7}$$

where $\mathbf{H}^i$ and $\mathbf{H}^r$ denote the incident and reflected magnetic field components at the surface, respectively, the current density on a locally flat surface simplifies to [1]:

$$\mathbf{J}_{\mathrm{s}} = 2\hat{\mathbf{n}} \times \mathbf{H}^i = 2\hat{\mathbf{n}} \times \mathbf{H}^r. \tag{2.8}$$

This approximation, known as the physical-optics approximation, holds when the transverse dimensions of the reflector, its radius of curvature, and the radius of the incident wavefront are all large compared to the wavelength [6]. In the far-field region, the current density can also be written in terms of the electric field components as:

$$\mathbf{J}_{\mathrm{s}} = \frac{2}{\eta}[\hat{\mathbf{n}} \times (\hat{\mathbf{s}}_i \times \mathbf{E}^i)], \tag{2.9}$$

or equivalently,

$$\mathbf{J}_{\mathrm{s}} = \frac{2}{\eta}[\hat{\mathbf{n}} \times (\hat{\mathbf{s}}_r \times \mathbf{E}^r)], \tag{2.10}$$

where η is the intrinsic impedance of the medium, $\hat{\mathbf{s}}_i$ and $\hat{\mathbf{s}}_r$ are unit vectors along the paths of the incident and reflected waves, and $\mathbf{E}^i$ and $\mathbf{E}^r$ are the corresponding electric fields [7].

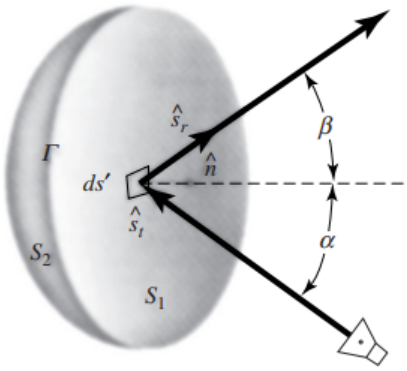

Figure 2.1: Reflecting surface with boundary Γ [1]

The scattered electromagnetic field is then determined by integrating these surface currents over the illuminated region. PO is an effective technique for predicting scattered fields from conductive bodies and is particularly useful in high-frequency scenarios, where only the local incident field is needed to calculate the induced surface currents.

The Method of Moments (MoM) is a well-known technique used to solve electromagnetic

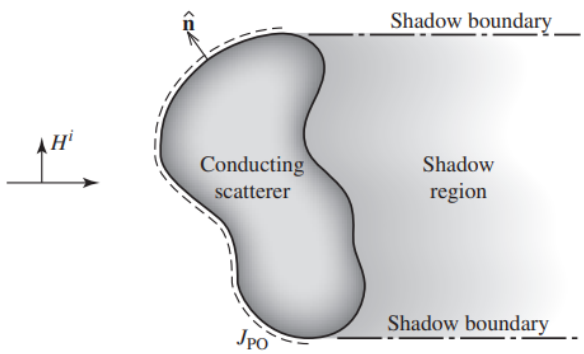

Figure 2.2: Physical optics current on a conducting scatterer [2]

(EM) field problems. It focuses on the integral equations derived from Maxwell's equations. The idea of changing a linear functional equation into a linear matrix equation has been studied since the 1800s, with Galerkin introducing his method in 1915. However, Galerkin's approach did not have a strong mathematical basis at first. Later, Pocklington created an integral equation and proposed a solution, but these methods were still difficult to use because they required a lot of complex calculations.
The MoM is now a popular computational method in electromagnetics, as it makes complicated problems easier to handle by turning them into matrix equations. This method is often used in software like FEKO. The two main parts of the MoM approach involve dividing the problem into smaller elements and using a basis function to approximate the current. This allows it to change boundary-value problems into matrix equations that can be solved by computers, making it especially useful for analyzing problems in open regions.

2.3. Method of Moments (MoM)

The MoM is a well-established full wave technique for solving electromagnetic (EM) field problems. It works by addressing the integral equation formulations of Maxwell's equations. The approach of converting a linear functional equation into a linear matrix equation has been under study since the 1800s, with Galerkin contributing his technique in 1915 [8]. However, Galerkin's method initially lacked a solid mathematical foundation [9]. Later, Pocklington developed an integral equation and suggested a solution [6]. Still, these methods remained impractical due to the extensive and expensive computations required.

The MoM is a widely used computational technique in electromagnetics that simplifies complex electromagnetic problems into manageable matrix equations. This method is commonly found in software like FEKO. The two key components of the MoM approach are breaking the problem into smaller elements and approximating the current using a basis function. It transforms boundary-value problems into matrix equations that can be

solved on computers, making it particularly useful for analyzing open-region problems such as antenna radiation and scattering.

For the MoM I used a bowtie antenna to illuminate an incident magnetic field to the PO $\lambda \times \lambda$ parabolic plate, the detailed information will be shown in the PO solver section. In MoM, the structure responsible for electromagnetic radiation or scattering is represented by equivalent surface currents, which are divided into smaller segments through meshing. By using Green's functions, a matrix equation is developed that describes the interactions between these segments, indicating how each one affects the others.

Solving this matrix equation provides an approximation of the current in each segment [6]. The size of the matrix depends on the problem's complexity and available computational resources [6]. In the MoM, the goal is to reduce the electromagnetic problem into a matrix equation form that can be solved numerically. The fundamental equation is typically written as:

$$[\mathbf{Z}][\mathbf{J}_{\mathrm{MoM}}] = [\mathbf{V}] \tag{2.11}$$

Here, $\mathbf{Z}$ is the impedance matrix, which describes how different segments of the structure interact with each other, while $\mathbf{J}_{\mathrm{MoM}}$ is the vector of unknown current densities on the structure that we need to solve for, and $\mathbf{V}$ is the known excitation vector, usually related to the incident fields. The relationship between these is given by:

$$\mathbf{J}_{\mathrm{MoM}} = [\mathbf{Z}^{-1}][\mathbf{V}] \tag{2.12}$$

Equation 2.12 shows how the surface current density is calculated in the MoM using the known excitation source $\mathbf{V}$ and the impedance matrix $\mathbf{Z}$, which is based on the antenna's mesh described in section 3.2.

2.4. Hybrid MoM/PO

The Hybrid MoM-PO approach is used to study how an antenna interacts with a larger structure in its vicinity. This method take advantage of MoM's precision for complicated shapes like antennas while combining it with the speed of PO for large, smooth surfaces like reflectors. MoM, which is a boundary element method, is known for its accuracy, but it can become too slow for large structures because of the large matrix equations involved. In contrast, PO is a high-frequency method that simplifies calculations but lacks accuracy for small details or close-range fields.

To benefit from both techniques, Hybrid MoM-PO method(which is already been

developed) is implemented by combine MoM and PO solutions [10]. These methods provide precise solutions for complex problems while maintaining efficiency. The hybrid MoM-PO method works by solving an equation to find the surface currents induced on the scatterer by an incoming electromagnetic wave, and then it calculates the scattered fields using these currents.

The solution of the matrix equation yields the current distribution across the bowtie antenna, which can then be used to compute the magnetic field $\mathbf{H}^{\mathbf{inc}}_{\mathbf{MoM}}(\mathbf{r})$ incident. To compute the total scattered magnetic field using a Hybrid MoM-PO, we follow the steps outlined below:

1. First, obtain the scattered magnetic field $\mathbf{H}^{\mathbf{MoM}}_{\mathbf{s}}(\mathbf{r})$ from MoM which is equal to $\mathbf{H}^{\mathbf{MoM}}_{\mathbf{i}}(\mathbf{r})$ for the Hybrid MoM-PO code:

$$\mathbf{H}^{\mathbf{MoM}}_{\mathbf{s}}(\mathbf{r}) = \mathbf{H}^{\mathbf{MoM}}_{\mathbf{i}}(\mathbf{r}) \tag{2.13}$$

2. Next, apply the following equation in the PO code to calculate the surface current density $\mathbf{J}_{\mathrm{PO}}$:

$$\mathbf{J}_{\mathrm{PO}} = 2(\hat{\mathbf{n}} \times \mathbf{H}^{\mathbf{inc}}_{\mathbf{MoM}}(\mathbf{r})) \tag{2.14}$$

3. Finally, use the calculated $\mathbf{J}_{PO}$ to determine the total scattered magnetic field $\mathbf{H}^{\mathbf{MoM}}_{\mathbf{s}}(\mathbf{r})$ with the equation:

2.5. Near-Field Radiation

The near-field region is the area close to an antenna or radiating structure where the field distribution is complex, and both electric ($\mathbf{E}$) and magnetic ($\mathbf{H}$) fields do not necessarily behave like plane waves. Instead, reactive components dominate [1]. In FEKO, near-field calculations focus on evaluating the electric ($\mathbf{E}$) and magnetic ($\mathbf{H}$) fields at a set of points or a grid close to the radiating structure [11].

Figure 2.3 illustrates the near-field coordinate system with a single observation point. The near-field calculation will be utilized in the following sections to present the results using PO, MoM, and a Hybrid MoM-PO approach. The results will first be shown for a single observation point, followed by plots over a grid of points to illustrate the field distribution as shown in figure 5.4. I will then compare these results with those from my MATLAB code, displaying both sets of plots side by side to verify if they match and identify any differences.

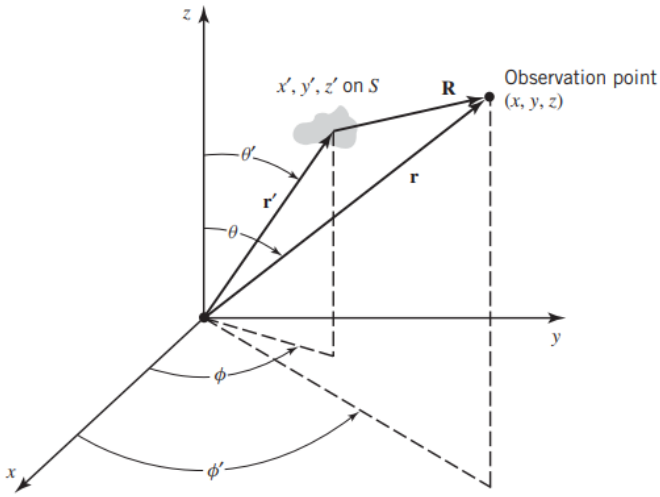

Figure 2.3: near-field coordinate system [1]

2.6. Far-Field Radiation

In antenna theory, the far-field region is the area where the characteristics of the radiated field stabilize and approximate a plane wave, with the field intensity decreasing inversely with distance. In this region, the electromagnetic fields behave as plane waves, maintaining a consistent shape and behavior over long distances. Typically, the far-field is considered to begin at a distance beyond $\frac{2\mathbf{D}^2}{\lambda}$, where $\mathbf{D}$ is the antenna's largest dimension and λ is the wavelength [7]. In FEKO simulations, the far-field region is often treated as if it extends infinitely.

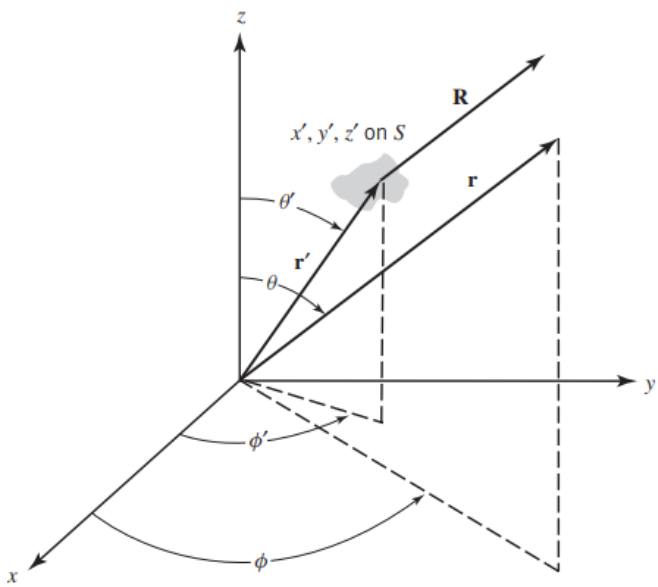

Figure 2.4: Far-field coordinate system [1]

Figure 2.4 illustrates a coordinate system with an observation point at a considerable distance from the radiation source. Initially, the fields radiated by an antenna are spherical,

but they gradually take on a plane wave form as they travel further from the source.

In the far-field, the complex magnetic field strength, $\tilde{H}_{\text{far}}$, is defined at this large distance R from the source [11]. This relationship can be expressed as:

$$\lim_{R\to\infty} \tilde{H}(\mathbf{r}) = \frac{e^{-j\beta_0 R}}{R}\tilde{H}_{\text{far}} \tag{2.16}$$

Where $\tilde{H}_{\text{far}}$ represents the far-field component, β_0 is the phase constant, and $R = |\mathbf{r}|$. In FEKO, these far-field values are computed using the angular components H_θ and H_ϕ as:

$$H_\theta = H_\theta^{\text{FF}} \frac{e^{-j\beta R}}{R} \tag{2.17}$$

$$H_\phi = H_\phi^{\text{FF}} \frac{e^{-j\beta R}}{R} \tag{2.18}$$

Where H_θ^{FF} and H_ϕ^{FF} are the far-field components in the θ and ϕ directions, respectively. These components are generally tabulated by their magnitude and phase, which allows for analysis of both amplitude and phase variation across the far-field.

For specific polarization requirements, the far-field components are described as follows:

- **S-polarization:**

$$\tilde{H}_{\text{far,S}} = \frac{1}{\sqrt{2}}(\tilde{H}_{\text{far},\phi} - \tilde{H}_{\text{far},\theta}) \tag{2.19}$$

- **Z-polarization:**

$$\tilde{H}_{\text{far,Z}} = \frac{1}{\sqrt{2}}(\tilde{H}_{\text{far},\phi} + \tilde{H}_{\text{far},\theta}) \tag{2.20}$$

In this context, $\tilde{H}_{\text{far},\theta}$ and $\tilde{H}_{\text{far},\phi}$ denote the magnetic field's magnitude and phase in the θ and ϕ orientations, respectively [11]. The S- and Z-polarization equations adjust these components to account for specific polarization directions in the far-field.

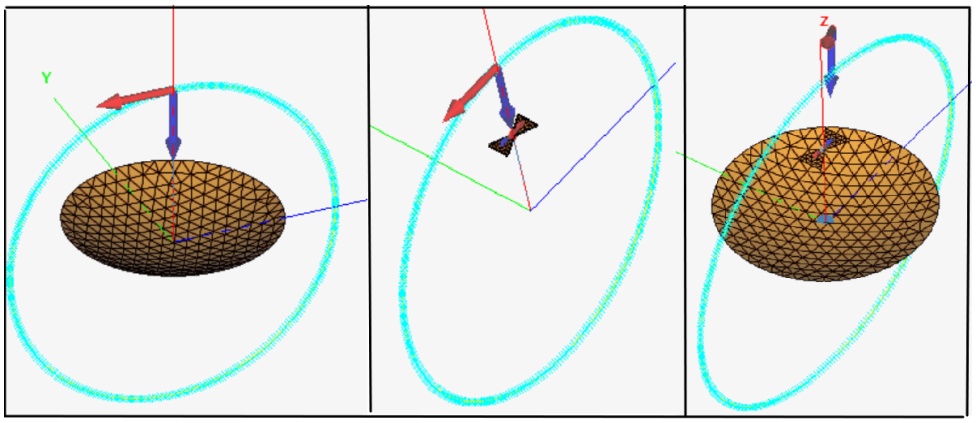

Figure 2.5: Far-field with 360 points around PO, MoM, and hybrid MoM-PO structures.

Figure 2.5 illustrates the far-field measurement points around different structures simulated using PO, MoM, and the hybrid MoM-PO approaches. In subsequent sections, results from MATLAB implementations of these methods will be compared with FEKO outputs, highlighting the alignment and differences between the computational approaches.

2.7. Shadowing Effect

The shadowing effect in electromagnetics occurs when parts of a reflector or conducting object are blocked from direct radiation, resulting in **"lit"** and **"shadowed"** regions on its surface [5].

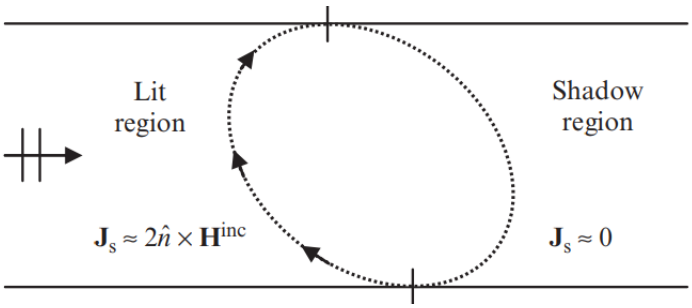

Figure 2.6: Illustration of Shadowing Effect.

In the **"lit region"** (facing the source of the incident field) in 2.6, an induced surface current density $\mathbf{J}_s$ forms, described by:

$$\mathbf{J}_s = 2\hat{\mathbf{n}} \times \mathbf{H}^i \tag{2.21}$$

Where $\hat{\mathbf{n}}$ is the unit normal vector to the surface, and $\mathbf{H}^i$ is the incident magnetic field. This relationship indicates that $\mathbf{J}_s$ depends on both the strength and direction of the incident field.

In the**"shadowed region"** (the opposite side of the reflector) in 2.6, there is no direct incident field, so the induced current is zero:

$$\mathbf{J}_s = 0 \tag{2.22}$$

The **"shadowed boundary"** divides the lit and shadowed regions and marks the edge where the incident field stops illuminating the surface. While accounting for shadowing is crucial for accurately predicting the induced surface currents and resulting radiation pattern, I did not include the shadowing effect in my MATLAB code due to time constraints

and the additional complexity required for region identification and current adjustments.

$$\mathbf{J}_s = \begin{cases} 2\hat{\mathbf{n}} \times \mathbf{H}^i & \text{on the front (lit) side of the reflector} \\ 0 & \text{on the shadowed side of the reflector} \end{cases} \tag{2.23}$$

2.8. Conclusion

This chapter provided an overview of essential electromagnetic methods, including PO, the MoM, and the Hybrid MoM-PO approach. It outlined how these techniques are applied to solve complex problems involving electromagnetic scattering, radiation from antennas, and interactions with large structures. The advantages and limitations of each method were highlighted, demonstrating how they complement each other for accurate and efficient analysis. Additionally, the chapter explained near-field and far-field concepts and the shadowing effect on reflectors.

Chapter 3

Physical Optics (PO) Solver

3.1. Magnetic Field Calculation

A Green's function represents the response of a system to a point source, and it is a fundamental solution to partial differential equations (PDEs) that describe physical phenomena. In electromagnetics, Green's function describes how an electromagnetic field propagates from a source to an observation point. For free-space, the Green's function for the scalar Helmholtz equation is used, and it is given by:

$$G_0(\mathbf{r}, \mathbf{r}') = \frac{e^{-jk_0|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|} \tag{3.1}$$

Where $\mathbf{r}$ and $\mathbf{r}'$ represent the observation and source points, respectively, $k_0 = \frac{2\pi}{\lambda}$ is the wave number, and λ is the wavelength. This Green's function describes the field at a distance $\mathbf{R} = |\mathbf{r} - \mathbf{r}'|$ from the source and assumes free-space propagation conditions.

3.1.1. Gradient of Green's Function

In the case of electromagnetic scattering problems like the one I'm solving now, the integral formulation involves Green's function and its gradient. The gradient of the Green's function, $\nabla G_0(\mathbf{r}, \mathbf{r}')$, plays a critical role in calculating the scattered fields. It can be expressed as:

$$\nabla G_0(\mathbf{r}, \mathbf{r}') = \left(jk_0 + \frac{1}{\mathbf{R}}\right) \frac{e^{-jk_0\mathbf{R}}}{4\pi\mathbf{R}} \bar{\mathbf{R}} \tag{3.2}$$

Where $\mathbf{R} = \mathbf{r} - \mathbf{r}'$ is the distance between the observation point and the source point, and $\bar{\mathbf{R}} = \frac{\mathbf{R}}{|R|}$ is the unit vector in the direction of $\mathbf{R}$.

3.1.2. Scattered Magnetic Field Calculation

With this formulation, the scattered magnetic field $\mathbf{H}_s(\mathbf{r})$ can be computed using the equivalent surface current distribution $\mathbf{J}(\mathbf{r}')$. The scattered magnetic field integral is given by:

$$\mathbf{H}_s(\mathbf{r}) = -\frac{1}{4\pi}\int_S \left(jk_0 + \frac{1}{\mathbf{R}}\right)\bar{\mathbf{R}} \times \mathbf{J}(\mathbf{r}')\frac{e^{-jk_0\mathbf{R}}}{4\pi\mathbf{R}}dS' \tag{3.3}$$

This integral formulation computes the scattered magnetic field by integrating the contributions from surface current densities $\mathbf{J}(\mathbf{r}')$ over the illuminated portion of the scattering object. The results of this equation 3.3 will be presented in the Results and Measurement section. I obtained them using MATLAB code and FEKO.

3.2. Discretization of the Current

Figure 3.1 shows a plate with triangular meshes, where the red dots denote the centroids of the triangular elements. This type of discretization divides the surface into smaller, manageable sections, aiding in numerical simulations of electromagnetic properties. The centroids are critical points where calculations for induced current density $\mathbf{J_{PO}}$ are performed.

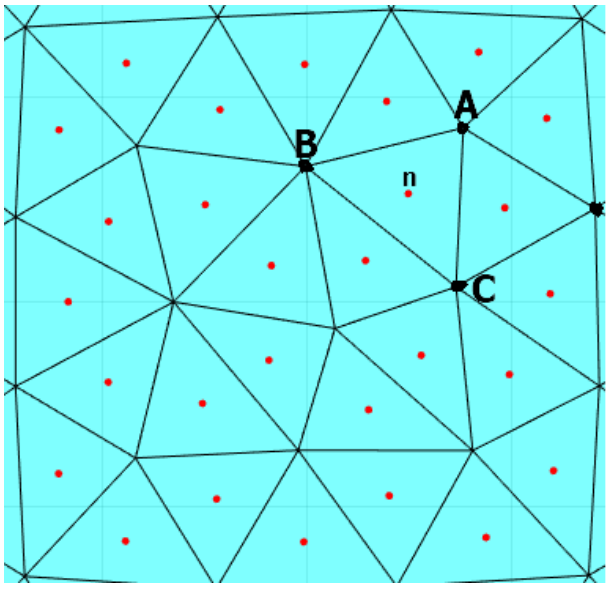

Figure 3.1: Meshed triangles with points A(x, y, z), B(x, y, z), and C(x, y, z)

Each triangular element is defined by its vertices $A(x, y, z)$, $B(x, y, z)$, and $C(x, y, z)$. These points are fundamental in calculating various geometric properties essential for

modeling the scattering problem. The centroid, for example, provides a reference point for calculating field interactions and induced currents.

The centroid of a triangle is given by:

$$\text{Centroid}(x, y, z) = \left(\frac{\mathbf{A}(x, y, z) + \mathbf{B}(x, y, z) + \mathbf{C}(x, y, z)}{3} \right) \tag{3.4}$$

The unit normal vector $\bar{\mathbf{n}}_i$ of each triangle is determined by taking the cross product of vectors **AB** and **AC**:

$$\bar{\mathbf{n}} = \mathbf{AB} \times \mathbf{AC} \tag{3.5}$$

This normal vector is normalized as follows:

$$\bar{\mathbf{n}}_i = \frac{\bar{\mathbf{n}}}{\|\bar{\mathbf{n}}\|} \tag{3.6}$$

The area Area_i of each triangular element is found using:

$$\text{Area}_i = \frac{1}{2} \|\mathbf{AB} \times \mathbf{AC}\| \tag{3.7}$$

The integral representing the total surface area, denoted in equation 4.5, is computed by summing the areas of all the triangular elements using MATLAB. The area is first computed over the vertices of each triangle before integrating over the entire mesh.

The unit normal vector $\bar{\mathbf{n}}$ is pivotal for calculating the current density:

$$\mathbf{J}_{PO} = 2(\bar{\mathbf{n}} \times \mathbf{H}^{\text{inc}}) \tag{3.8}$$

This current density plays a significant role in determining the scattered magnetic field $\mathbf{H}_s(\mathbf{r})$, as shown in the equations below.

$$\mathbf{H}_{\text{s}}(\mathbf{r}) = \int \int_S \mathbf{J}_{PO}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}')\, dS' \tag{3.9}$$

$$\mathbf{H}_{\text{s}}(\mathbf{r}) = \sum_i \mathbf{J}_{PO}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}'_{\mathbf{i}}) \times Area_i \tag{3.10}$$

The same approach of discretization of the current will be applied in chapter 4 and 5 where the same equation 4.5 will be used to calculate the scattered magnetic field $\mathbf{H}_{\text{s}}(\mathbf{r})$. The PO method was applied to analyze a perfectly conducting parabolic plate measuring $1\lambda \times 1\lambda$ (which is just a test I used in FEKO for the reflector) shown below 3.2.

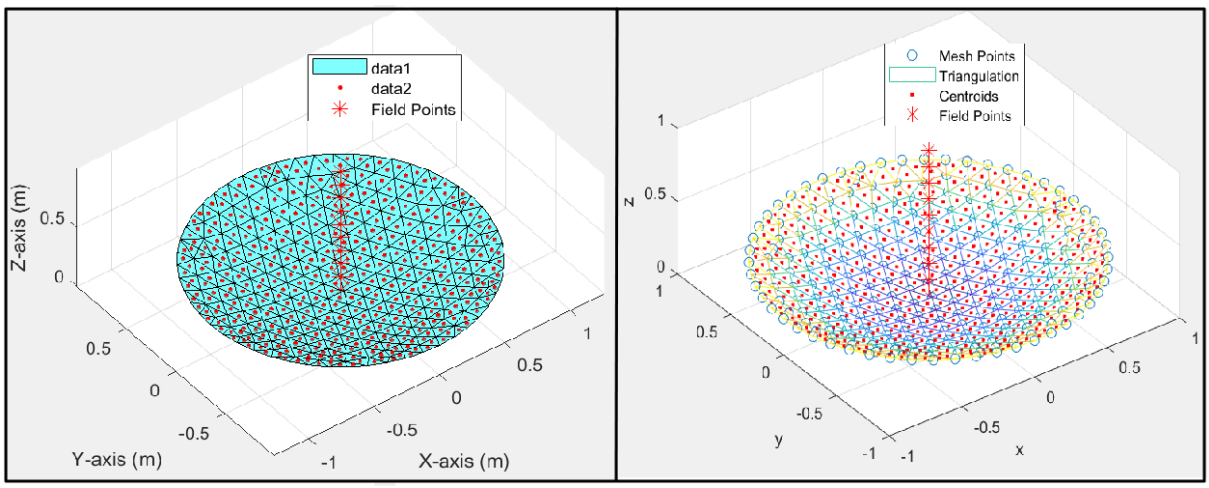

Figure 3.2: Mesh Visualization for a $1\lambda \times 1\lambda$

The mesh resolution was set to $\lambda/10$ for each triangular edge length and a frequency of $f = 300 \times 10^6$ so that accurate results in calculating the scattered magnetic field could be obtained. As shown in table 3.1, the calculated currents on the parabolic plate, subjected to a normally incident plane wave, closely matched the reference results obtained from FEKO.

3.3. PO for Plane Wave Scattering

This section explains the process of calculating the scattered magnetic field $\mathbf{H}_s$ using PO in MATLAB, along with corresponding FEKO simulations. In order to calculate the physical optics current $\mathbf{J}_{\text{PO}} = 2(\bar{\mathbf{n}} \times \mathbf{H}^i)$ you need to calculate the induced magnetic field $\mathbf{H}^i$ at each centroid of the mesh [12]. It is calculated based on a meshed rectangular plate with the PO using the information described in section 3.2, an example of that plate is shown in figure 3.3.

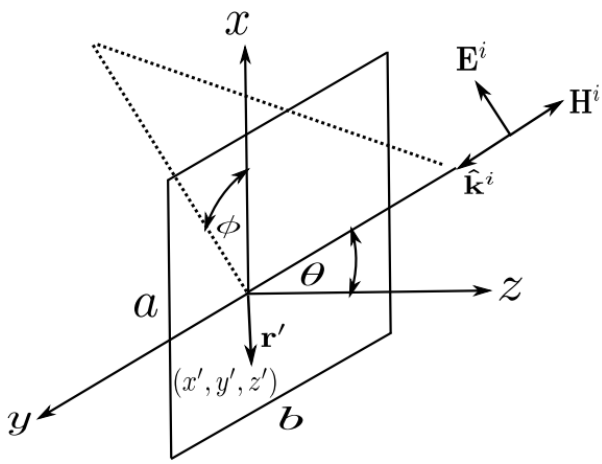

Figure 3.3: Rectangular plate with incident plane wave

The vector $\mathbf{E}^i$ represents the planar incident electric field with arbitrary polarization, arriving at an angle (θ, ϕ). The polarization of the wave is specified by the constants E_θ and E_ϕ, as given by:

$$\mathbf{E}^i = \left(E_\theta \bar{\theta} + E_\phi \bar{\phi}\right) e^{-j\mathbf{k}\cdot\mathbf{r}} \quad (V/m) \tag{3.11}$$

If we define the wavenumber vector as:

$$\bar{\mathbf{k}} = \mathbf{a_x} k_x + \mathbf{a_y} k_y + \mathbf{a_z} k_z = k\bar{\mathbf{a}}_\mathbf{n} \tag{3.12}$$

Expressing $\bar{\mathbf{a}}_\mathbf{n}$ in terms of the spherical angles ϕ and θ, we have:

$$\bar{\mathbf{a}}_\mathbf{n} = \sin\theta\cos\phi\,\bar{x} + \sin\theta\sin\phi\,\bar{y} + \cos\theta\,\bar{z} \tag{3.13}$$

Thus, the equation for $\mathbf{k}$ becomes:

$$\bar{\mathbf{k}} = k(\sin\theta\cos\phi\,\bar{x} + \sin\theta\sin\phi\,\bar{y} + \cos\theta\,\bar{z}) \tag{3.14}$$

and a radius vector from the origin:

$$\bar{\mathbf{r}} = \mathbf{a_x} x + \mathbf{a_y} y + \mathbf{a_z} z \tag{3.15}$$

When the wave travels toward the origin, the propagation vector $\bar{\mathbf{k}}$ and $\bar{\mathbf{r}}$ is the unit vector pointing to the incident field location. The magnetic field intensity can then be expressed as:

$$\mathbf{H}^i = -\frac{1}{\eta}\left(\bar{\mathbf{a}}_n \times \mathbf{E}^i\right) e^{-jk_n\cdot\mathbf{r}} \quad (\mathrm{A/m}) \tag{3.16}$$

It is now evident that a uniform plane wave propagating in an arbitrary direction $\bar{\mathbf{a}}_\mathbf{n}$ is a transverse electromagnetic (TEM) wave, with $\mathbf{E} \perp \mathbf{H}$ and that both $\mathbf{E}$ and $\mathbf{H}$ are normal to the direction of propagation $\bar{\mathbf{a}}_\mathbf{n}$. Using the equation 3.16 I was able to calculate the induced current density $\mathbf{J}_{\mathrm{PO}} = 2(\bar{\mathbf{n}} \times \mathbf{H}^i)$ at each centroid of my meshed plate.

The next step was to calculate the scattered magnetic field and the equation is given by [2]:

$$\bar{\mathbf{H}}_s(\mathbf{r}) - \mathcal{L}(\mathbf{M}_s) - \mathcal{K}(\bar{\mathbf{J}}_s) = \begin{cases} \bar{\mathbf{H}}(\mathbf{r}), & \mathbf{r} \in V_\infty \\ 0, & \mathbf{r} \in V_o \end{cases} \tag{3.17}$$

The magnetization term $\mathcal{L}(\mathbf{M}_s)$ is zero because we are working with conducting material, not dielectric material. To calculate the scattered field, you need to make use of the

polarization angles (θ and ϕ). As a result, the calculation formula for the scattered magnetic field will be as follows:

$$\bar{\mathbf{H}}_s(\mathbf{r}) = -\mathcal{K}(\bar{\mathbf{J}}_s) \tag{3.18}$$

The provided equations can be implemented in MATLAB to compute the scattered magnetic field across each 3D triangular element of the mesh. Additionally, the results obtained in MATLAB can be compared with those from FEKO to validate the accuracy and consistency of the calculations. Figure 3.4 shows a pseudo-code of how the PO is implemented in MATLAB.

(a) **Import the Mesh:** Load the mesh geometry from a NASTRAN file.

(b) Enter the θ and ϕ values for the plane wave.

(c) **Calculate Centroids and Normals:** Compute the centroid and normal vector $\bar{\mathbf{n}}$ for each triangle.

(d) **Incident Electric Field Calculation:** Evaluate the incident electric field $\mathbf{E}^i$ at each centroid:

$$\mathbf{E}^i = \left(E_\theta \bar{\theta}_0 + E_\phi \bar{\phi}_0\right) e^{-j\mathbf{k}\cdot\mathbf{r}}. \tag{3.19}$$

(e) **Incident Magnetic Field Calculation:** Derive the incident magnetic field $\mathbf{H}^i$:

$$\mathbf{H}^i = -\frac{1}{\eta}\left(\bar{\mathbf{a}}_n \times \mathbf{E}^i\right) e^{-j\mathbf{k}\cdot\mathbf{r}}. \tag{3.20}$$

(f) **Induced Current Density:** Calculate the induced surface current density $\mathbf{J}_{\mathrm{PO}}$:

$$\mathbf{J}_{\mathrm{PO}} = 2(\bar{\mathbf{n}} \times \mathbf{H}^i). \tag{3.21}$$

(g) Calculate the gradient of Green's Function using the equation below with $\mathbf{R} = |\mathbf{r} - \mathbf{r}'|$:

$$\nabla G_0(\mathbf{r}, \mathbf{r}') = \frac{(\mathbf{r} - \mathbf{r}')\,(1 + jk_0\mathbf{R})}{4\pi\mathbf{R}^3} e^{-jk_0\mathbf{R}} \tag{3.22}$$

(h) **Scattered Magnetic Field:** Compute the scattered magnetic field $\mathbf{H}_s(\mathbf{r})$ using Green's function:

$$\mathbf{H}_s(\mathbf{r}) = \int\int_S \mathbf{J}_{\mathrm{PO}}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}') dS'. \tag{3.23}$$

$$\mathbf{H}_{\mathrm{s}}(\mathbf{r}) = \sum_i \mathbf{J}_{PO}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}'_{\mathbf{i}}) \times Area_i \tag{3.24}$$

(i) Plot the near-field and Far-field results of the Physical Optics (PO) comparing MATLAB and FEKO outcomes.

Figure 3.4: Pseudo code for the basic PO implementation in POsolver.m

The process of calculating the scattered magnetic field using PO involves several key computational steps. First, the object's geometry is imported from a mesh file, and the centroids and normal vectors of the triangular mesh elements are computed using the information in section 3.2, you then have to solve the PO surface current density $\mathbf{J}_{PO}$ at the mesh element centroids then the incident electric and magnetic fields are evaluated at each centroid [13], and finally, the scattered magnetic field is obtained using Green's function. These steps are summarized in Figure 3.4.

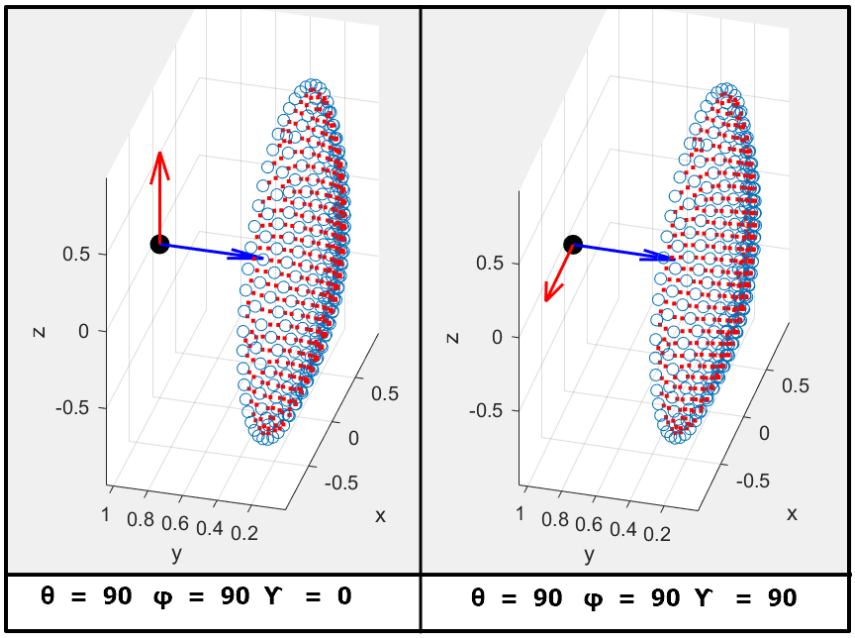

Figure 3.5: Plane Wave Source with different θ, ϕ, and γ values

Figure 3.5 illustrates a parabolic plate illuminated by a plane wave source at various angles denoted as θ and ϕ, indicated by the blue arrow. Additionally, γ is used to control the direction of the red arrow, which represents the incident electric field $\mathbf{E}^i$. In this setup, I used a plane wave source in FEKO to test the response of the parabolic plate to various incident angles. The plane wave's direction is controlled by the angles θ and ϕ, allowing for the calculation of the incident electric field $\mathbf{E}^i$ at each triangle centroid, as defined in equation 3.19.

Table 3.1: Comparison of the magnitude of $|\mathbf{J}_{PO}|$ and H-Field in (mA/m) at various angles using FEKO and MATLAB simulations.

Angle θ and ϕ	FEKO		MATLAB	
	$\lvert\mathbf{J}_{\mathrm{PO}}\rvert[mA/m]$	$\lvert\mathbf{H}_s\rvert[mA/m]$	$\lvert\mathbf{J}_{\mathrm{PO}}\rvert[mA/m]$	$\lvert\mathbf{H}_s[mA/m]\rvert$
0°	5.3401	7.153	5.309	6.853
30°	5.400	2.065	5.3089	1.891
45°	5.391	3.281	5.391	3.133
60°	5.392	3.367	5.391	3.321
90°	5.401	3.601	5.384	3.582

Table 3.1 presents the results for the magnitude of the current surface density, denoted as $\mathbf{J}_{\mathrm{PO}}$, which was calculated using the PO method and the scattered magnetic field $\mathbf{H}_s(\mathbf{r})$ at various angles of the plane wave, controlled by the angles θ and ϕ, as illustrated in Figure 3.5. The purpose here is to compare FEKO results with MATLAB to verify the consistency between the two methods. As can be seen, the numbers for $\mathbf{J}_{\mathrm{PO}}$ and $\mathbf{H}_s(\mathbf{r})$ at various angles are pretty close. This suggests that both approaches produce similar results, which indicates that the PO formulation is correctly implemented in MATLAB.

My results in MATLAB were approximately 0.3% lower than those of FEKO, and this is because MATLAB's numerical methods may introduce slight differences due to how the magnetic field integral is evaluated (the MATLAB code assumes constant current over each triangle). Additionally, FEKO may use more advanced algorithms or refined settings by default, leading to slightly higher precision in its results.

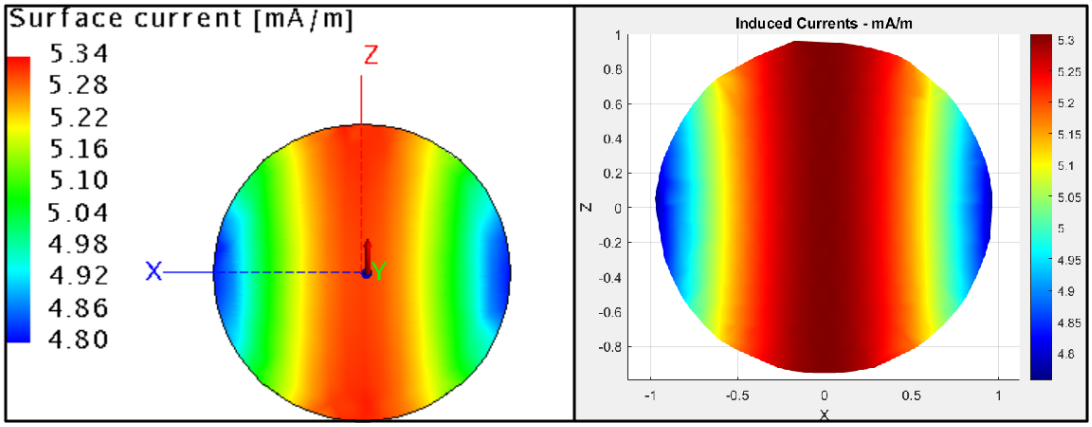

Figure 3.6: Image of the Surface current [mA/m] density of FEKO and MATLAB

The figure 3.6 above displays two images. The left image shows the results of the surface current $\mathbf{J}_{\mathrm{PO}}$ as represented in the color bar in FEKO, while the right image presents the MATLAB results for $\theta = 90°$ and $\phi = 90°$. This figure illustrates how the plane wave source illuminates the parabolic plate and demonstrates the distribution of current on the plate. As observed, both colour bars indicate similar results, confirming that my MATLAB code implementation aligns with the results obtained in FEKO.

3.4. Near-Field Results

This section presents the near-field results at different observation points. First, I'll show the result for a single observation point, then expand to results over multiple points, like those in Figure 5.4. Utilizing this set of observation points to compute the scattered magnetic field, $\mathbf{H}^s(\mathbf{r})$ at various locations. In FEKO, near-field values can be computed in either Cartesian or spherical coordinates.

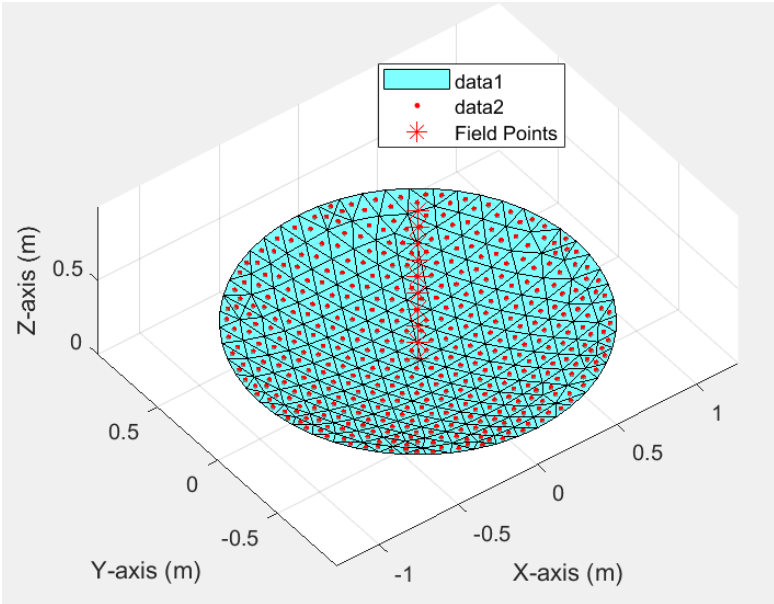

Figure 3.7: 10 observation points

The table 3.2 below gives the magnitude results of the scattered magnetic $\mathbf{H}_s(\mathbf{r})$ field at different observation points along the z-axis computed from FEKO and MATLAB. The coordinates are given as $[x, y, z]$ where x and y are fixed at zero and z is varied, with the incident angle of $\theta = 0°$ and $\phi = 0°$. These results are clearly shown as a graph shown in Figure 3.9.

Table 3.2: The table shows the magnitude of the H-Field results at various observation points (5 points) using FEKO and MATLAB simulations.

Observation Points [x,y,z]	**FEKO**	**MATLAB**
	H – Field [A/m]	
[0, 0, 0]	0.7173	0.653
[0, 0, 0.25]	4.591	4.8544
[0, 0, 0.50]	7.187	6.944
[0, 0, 0.75]	7.432	7.321
[0, 0, 1.0]	8.214	8.102

From the result, one can observe that FEKO and MATLAB give close values of the scattered magnetic field $\mathbf{H}_s(\mathbf{r})$ where in most of the incidences FEKO is a little higher. The greater the z-axis distance from the origin, the closer the results provided by both methods, testifying to great concordance between the two methods in the case of points further way from the surface.

For this project, I chose Cartesian coordinates to obtain a set of points along a line. The software calculates the near-field by numerically solving Maxwell's equations over the

meshed structure, using the following Pseudo Code:

(a) **Initialize Field Points:**

$\Rightarrow$ Start_Point: [0, 0, 0]
$\Rightarrow$ End_Point: [0, 0, λ]
$\Rightarrow$ Increment: [0, 0, 0.2] (distance between points)
$\Rightarrow$ Number of Field Points: [0, 0, 10], as shown in Figure 5.4.

(b) Generate field points along each axis (x, y, z).

(c) Create a grid of field points as shown in Figure 5.4.

(d) Use the field points as observation points ($\mathbf{r} \approx \mathbf{r}_{\text{observer}}$ in my MATLAB code), then calculate $\mathbf{R} = |\mathbf{r} - \mathbf{r}'|$.

(e) Calculate the gradient of Green's Function using the equation below with $\mathbf{R} = |\mathbf{r} - \mathbf{r}'|$:

$$\nabla G_0(\mathbf{r}, \mathbf{r}') = \frac{(\mathbf{r} - \mathbf{r}')\,(1 + jk_0\mathbf{R})}{4\pi\mathbf{R}^3} e^{-jk_0\mathbf{R}} \tag{3.25}$$

(f) **Scattered Magnetic Field:** Compute the scattered magnetic field $\mathbf{H}_s(\mathbf{r})$ using Green's function:

$$\mathbf{H}_s(\mathbf{r}) = \int\int_S \mathbf{J}_{\text{PO}}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}') dS'. \tag{3.26}$$

$$\mathbf{H}_{\text{s}}(\mathbf{r}) = \sum_i \mathbf{J}_{PO}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}'_{\mathbf{i}}) \times Area_i \tag{3.27}$$

(g) Plot the near-field results for PO, comparing the outcomes from FEKO and MATLAB in a single graph.

Figure 3.8: Pseudo-code Near-Field implimentation

Figure 3.9 above shows the near-field plot of the scattered magnetic field $\mathbf{H}_s(\mathbf{r})$ with 10 observation points (V position (m)), comparing the results obtained from FEKO with those from MATLAB. The plot demonstrates that the near-field results from both FEKO and MATLAB align very closely.

This figure highlights the consistency between the FEKO and MATLAB results in Cartesian coordinates, indicating accurate alignment across the observation points.

This alignment validates the numerical and integral methods used for the near-field calculations, as detailed in the pseudocode shown in Figure 3.8. It also confirms that the near-field data imported from FEKO into MATLAB matches the computed values in my MATLAB code.

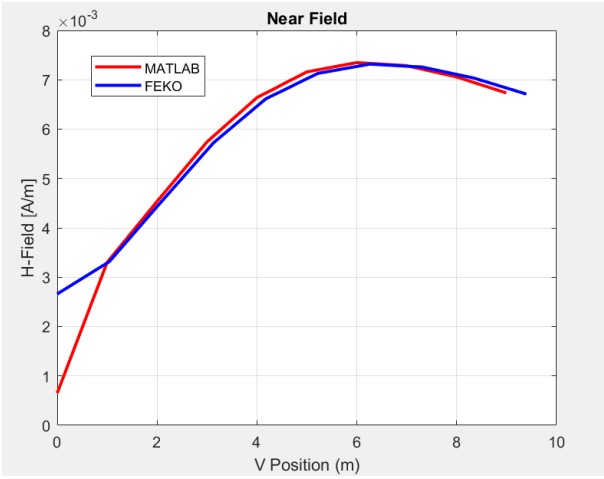

Figure 3.9: Near Field plot for both MATLAB and FEKO

3.5. Far-Field Results

Figure 5.6 compares far-field results computed using MATLAB and FEKO, illustrating the far-field **E**-field magnitude as a function of the angle θ (measured in degrees) while fixing $\phi = 0$.

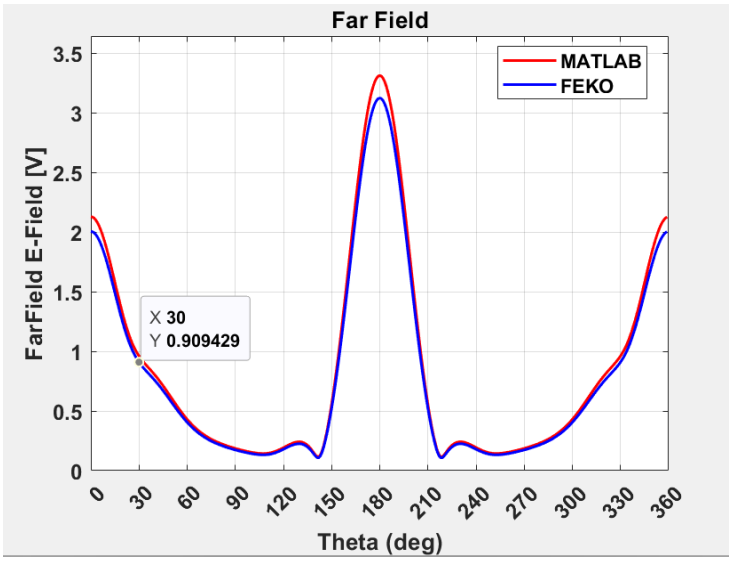

Figure 3.10: Far Field plot for both MATLAB and FEKO

The figure above shows two lines representing the results of the scattered $\mathbf{E}_s$ field the one in red is the results obtained in MATLAB and another in blue results imported from FEKO. These lines align closely across the angular range, indicating that both methods yield similar results.

A key observation is that a peak value occurs at $\theta = 180°$, suggesting a strong radiation pattern in this direction. This plot represents the radiation pattern of a scattering object

(parabolic plate), detailing how the **E**-field varies with direction in the far-field region. The plot shows θ ranging from 0° to 360° on the x-axis, while the y-axis displays the E-field magnitude in (V/m).

Figure 3.11 will include the pseudocode for the far-field calculations in MATLAB, highlighting the main structure of the algorithm and its key steps.

(a) Initialize the magnetic field $\mathbf{H}_s(\mathbf{r})$ array with 3 components (x, y, z) for each observation point.

(b) Loop over θ angles for spherical observation points from 0 to 360 degrees while fixing ϕ at 0.

(c) Inside the loop, convert spherical coordinates to Cartesian coordinates for far-field observation points.

(d) Multiply the Cartesian coordinates by the far-field distance of $\mathbf{r}_{FF} \approx 1 \times 10^6 \lambda$.

(e) Use the far-field points as observation points ($\mathbf{r} \approx \mathbf{r}_{\mathrm{FF}}$ in my MATLAB code), then calculate $\mathbf{R} = |\mathbf{r} - \mathbf{r}'|$.

(f) Calculate the gradient of Green's Function using the equation below with $\mathbf{R} = |\mathbf{r} - \mathbf{r}'|$:

$$\nabla G_0(\mathbf{r}, \mathbf{r}') = \frac{(\mathbf{r} - \mathbf{r}')\,(1 + jk_0\mathbf{R})}{4\pi\mathbf{R}^3} e^{-jk_0\mathbf{R}} \tag{3.28}$$

(g) **Scattered Magnetic Field:** Compute the scattered magnetic field $\mathbf{H}_s(\mathbf{r})$ using Green's function:

$$\mathbf{H}_s(\mathbf{r}) = \int\int_S \mathbf{J}_{\mathrm{PO}}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}') dS'. \tag{3.29}$$

$$\mathbf{H}_{\mathrm{s}}(\mathbf{r}) = \sum_i \mathbf{J}_{PO}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}'_{\mathbf{i}}) \times Area_i \tag{3.30}$$

(h) Convert the $\mathbf{H}_s(\mathbf{r})$ to $\mathbf{E}_s(\mathbf{r})$, where $\mathbf{r}$ is a function of θ and ϕ. The conversion is as follows [14]:

$$\mathbf{H}_s(r) = \frac{1}{\eta}(\mathbf{r} \times \mathbf{E}_s) \tag{3.31}$$

$$\eta(\mathbf{r} \times \mathbf{H}_s) = \mathbf{r} \times (\mathbf{r} \times \mathbf{E}_s) \tag{3.32}$$

(i) Plot the far-field results for PO comparing the outcomes from FEKO and MATLAB in a single graph 5.6.

Figure 3.11: Pseudo-code for Far-Field Implementation

Figure 3.12 below compares the far-field radiation patterns of an antenna or scattering

object, with results from MATLAB on the left and FEKO on the right.

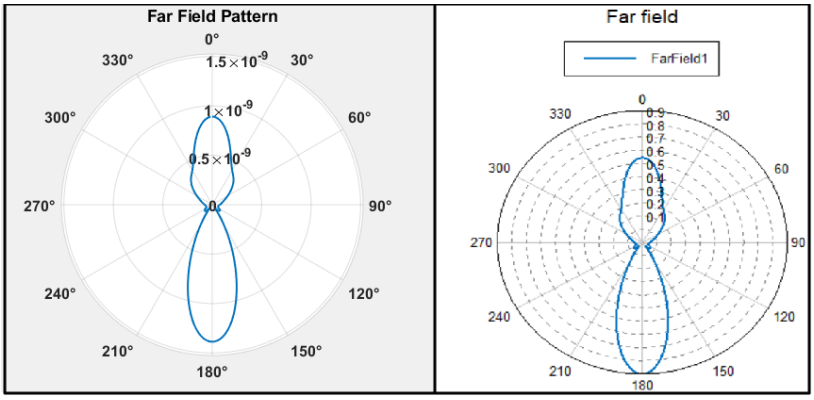

Figure 3.12: Far Field plot for both MATLAB and FEKO in Polar plot

Using polar plots, both plots display the far-field electric field (E-field) as a function of the angle θ (in degrees). The patterns are similar in shape, showing a symmetric radiation pattern with peak values around $\theta = 180°$. The similarity in these plots indicates that both MATLAB and FEKO have the same shape.

3.6. Conlcusion

This chapter focused on the development and implementation of a PO solver, covering both theoretical concepts and practical steps for calculating scattered magnetic fields using Green's function and its gradient. Accurate centroid and normal vector calculations for triangular mesh elements were highlighted.

Near-field and far-field analyses showed strong agreement between MATLAB and FEKO results, with minor near-origin discrepancies resolving at a distance. Far-field results were consistent, featuring matching E-field magnitudes and a peak at $\theta = 180°$. Pseudocode and visual plots validated the solver's accuracy, confirming MATLAB's reliability for electromagnetic analysis.

Chapter 4

Method of Moments (MoM) solver

4.1. Method of Moment (MoM)

For the MoM, I utilized a bowtie antenna with an approximate length of $\frac{\lambda}{2}$. The initial code was provided by my supervisor, and I made modifications to include the calculation of the radiated magnetic field, $\mathbf{H_s^{MoM}}(\mathbf{r})$. This field serves as the illuminating source for my Physical Optics (PO) implementation, forming the basis of my hybrid MoM-PO approach, which is described in detail in the subsequent chapter 5.

Figure 4.1 below displays the meshed bowtie antenna that I utilized in my code to calculate the radiated magnetic field $\mathbf{H_s^{MoM}}(\mathbf{r})$, the current density $\mathbf{J}_{\text{MoM}}$, and the Green's function $G_0(\mathbf{r}, \mathbf{r}')$ using. The normal vector and centroid values of the antenna's triangular mesh will be calculated using the information provided in section 3.2, which has already been explained. Additionally, I used this antenna for plotting the near-field and far-field results which will be shown in this chapter.

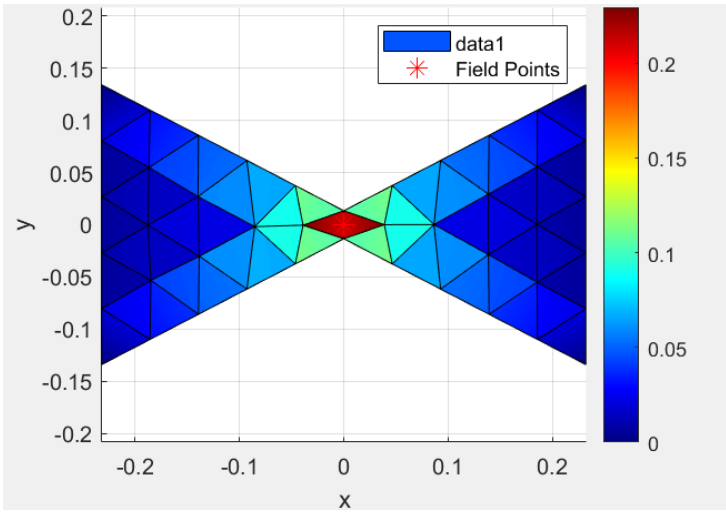

Figure 4.1: Mesh Visualization for a Bowtie antenna

In the MoM, the process begins by initializing the impedance matrix $[Z]$, which captures the interaction between basis functions using Green's functions. A voltage source $[V]$ is

then applied as excitation in the bowtie antenna in FEKO and MATLAB. The current density $\mathbf{J}_{\text{MoM}}$ is calculated by solving the matrix equation:

$$\mathbf{J}_{\text{MoM}} = [\mathbf{Z}^{-1}][\mathbf{V}] \tag{4.1}$$

Finally, the radiated magnetic field $\mathbf{H_s^{MoM}}(\mathbf{r})$ is computed by integrating the current density and the gradient of the Green's function over the surface. This method effectively models how electromagnetic fields interact with conducting structures.

Figure 4.2 shows the pseudo-code of the MoM, the explanation of each part of the process provided below:

(a) **Import the Mesh:** Load the bowtie antenna's mesh geometry from a NASTRAN file.

(b) Input the θ and ϕ values for the plane wave excitation.

(c) **Calculate Centroids and Normals:** Compute the centroid and normal vector $\hat{\mathbf{n}}$ for each triangle.

(d) **Initialize the Impedance Matrix:** Set up the impedance matrix $\mathbf{Z}$ using the provided MoM library.

(e) **Initialize the Voltage Source:** Define the voltage gap source $[\mathbf{V}]$ (edge port in FEKO) for excitation.

(f) **Compute the Current Density:** Solve for the current density $\mathbf{J}_{\text{MoM}}$ using the inverse impedance matrix and the voltage source.

$$\mathbf{J}_{\text{MoM}} = [\mathbf{Z}]^{-1}[\mathbf{V}] \tag{4.2}$$

(g) Calculate the Green's Function using the equation below:

(h) Calculate the gradient of Green's Function using the equation below with $\mathbf{R} = |\mathbf{r} - \mathbf{r}'|$:

$$\nabla G_0(\mathbf{r}, \mathbf{r}') = \frac{(\mathbf{r} - \mathbf{r}')\,(1 + jk_0\mathbf{R})}{4\pi\mathbf{R}^3} e^{-jk_0\mathbf{R}} \tag{4.3}$$

(i) **Scattered Magnetic Field:** Compute the scattered magnetic field $\mathbf{H}_s(\mathbf{r})$ using Green's function:

$$\mathbf{H_s^{MoM}}(\mathbf{r}) = \int\int_S \mathbf{J}_{\text{MoM}}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}') dS'. \tag{4.4}$$

$$\mathbf{H_s^{MoM}}(\mathbf{r}) = \sum_i \mathbf{J}_{MoM}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r_i'}) \times Area_i \tag{4.5}$$

(j) Plot the near-field and far-field results of the MoM comparing MATLAB and FEKO outcomes.

Figure 4.2: Pseudo code for the basic MoM implementation in `MoMsolver.m`.

To calculate the current density $\mathbf{J}_{\text{MoM}}$ at the centroids of the mesh of the bowtie antenna, I employ the same approach as outlined in the Physical Optics section 3.2. This involves using points A(x, y, z), B(x, y, z), and C(x, y, z) the vertices of the triangles in the mesh to determine the centroid of each triangle, the normal vector $\hat{\mathbf{n}}$, and the area of each triangle, as described in Equation 3.7. For this section, I used the MoM solution current at each mesh element centroid, not the PO approximation.

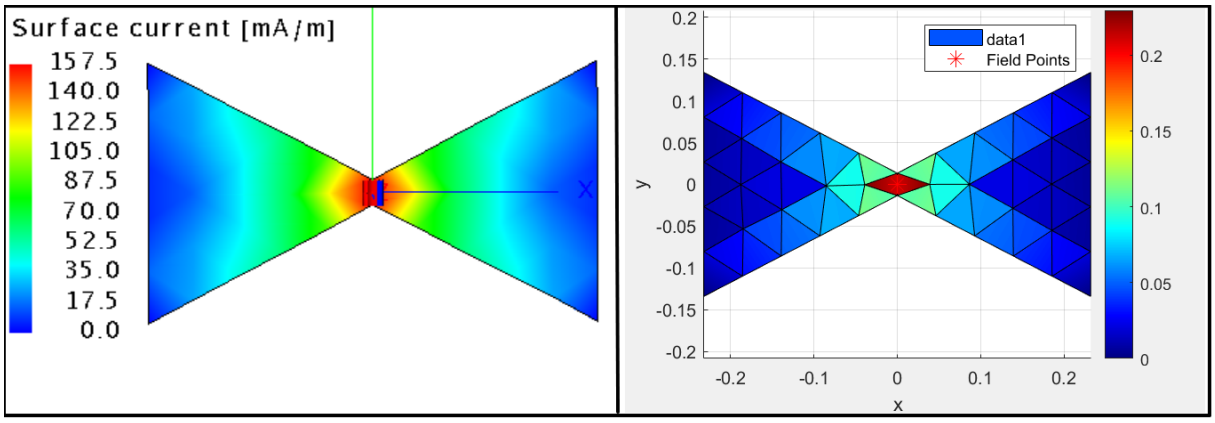

Figure 4.3: Surface current $\mathbf{J}_{\text{MoM}}$ density of FEKO and MATLAB

Figure 4.3 above shows the surface current density $\mathbf{J}_{\text{MoM}}$ distributed on the bowtie antenna that is excited by a voltage source $V = 1.0\angle 0$V in FEKO and MATLAB. The left image represents the results from FEKO, while the right image displays the results from MATLAB. Both visualizations include a color bar that represents the current density in units of [mA/m] for the FEKO one and [A/m] for the MATLAB one, with consistent color scaling between the two plots. The figures highlight the region where the voltage source is placed on the bowtie antenna, showing similar current distribution patterns across both images.

Table 4.1: Comparison of the magnitude of $\mathbf{J}_{\text{MoM}}$ and H-Field in (mA/m) at different excitation voltage [V] using FEKO and MATLAB simulations.

Excitation Voltage [V]	FEKO		MATLAB	
	$\lvert\mathbf{J}_{\text{MoM}}\rvert[mA/m]$	$\lvert\mathbf{H}_s\rvert[mA/m]$	$\lvert\mathbf{J}_{\text{MoM}}\rvert[mA/m]$	$\lvert\mathbf{H}_s\rvert[mA/m]$
$V = 1.0\angle 0$	157.5	112.5	156.41	112.04
$V = 0.8\angle 0$	150.0	90.0	147.3	89.63
$V = 0.6\angle 0$	112.5	75.0	109.7	73.2
$V = 0.4\angle 0$	75.0	45.0	67.4	44.81
$V = 0.2\angle 0$	32.0	22.5	31.28	22.4
$V = 0.0\angle 0$	0	0	0	0

Table 4.1 presents the results of the surface current density $\mathbf{J}_{\text{MoM}}$ calculated using the MoM, along with the scattered magnetic field $\mathbf{H}^{\text{inc}}_{\text{MoM}}(\mathbf{r})$ for different excitation voltage [V]. The table clearly shows that the MoM results align well with those obtained using FEKO.

4.2. Near-Field Results

In this section, I will present the near-field results at a defined set of observation points. Figure 4.4 below illustrates a meshed representation of a bowtie antenna, alongside a series of field observation points aligned along the z-axis.

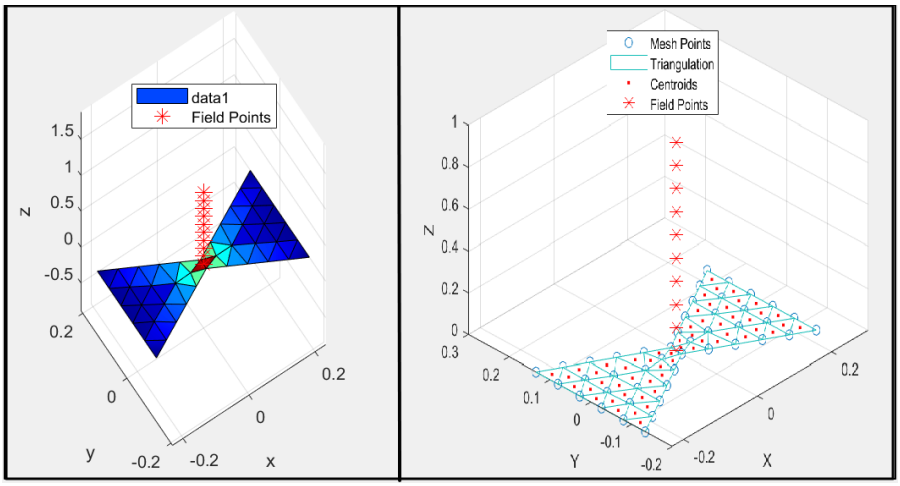

Figure 4.4: Bowtie antenna with 10 sets of observation points

These observation points are spaced uniformly, ranging from $z = 0$ to $z = 1$ meters, with an increment of 0.2 meters between each point. This setup allows for a detailed analysis of the field distribution along this axis.

In Figure 4.5, the results of the scattered magnetic field, represented as the **H**-field, are shown as a function of the observation points along the z-axis. The comparison includes data obtained from both MATLAB and FEKO, providing insight into the behavior of the near-field closer to the antenna.

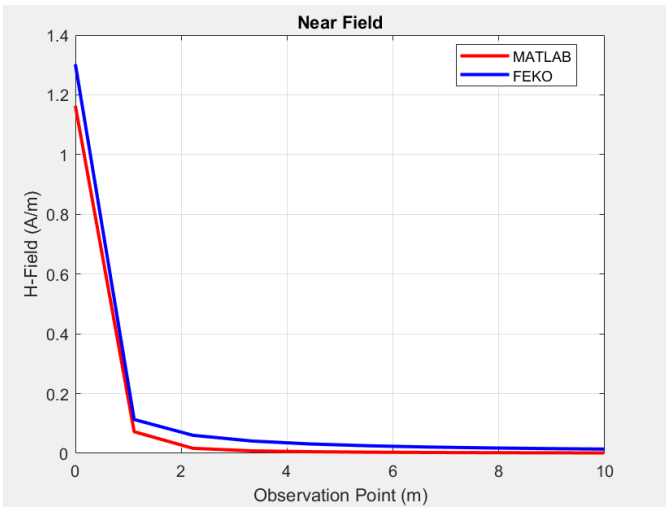

Figure 4.5: Near Field plot for both MATLAB and FEKO for the MoM part

Figure 4.5 above demonstrates that the near-field results obtained from the FEKO simulation closely match those generated using my MATLAB code. This strong alignment between the two sets of results validates the accuracy of the integral methods implemented in MATLAB for calculating the near-field behavior. The consistency between the MATLAB and FEKO results indicates that the computational approach used in MATLAB, as outlined in the pseudocode 3.8 in Chapter 2, is effective and reliable for near-field analysis.

4.3. Far-Field Results

This section presents the far-field results at the observation points. In the far-field region, the **E**-field and **H**-field propagate perpendicularly to each other and remain in phase, forming a plane wave. Both the **E**-field and **H**-field decay at a rate of $\frac{1}{r}$, where **r** represents the distance from the source.

Typically, the far-field region is defined for distances greater than $\frac{\lambda}{2\pi}$. The radial distance **R** from any point on the source or scatterer to the observation point can be aligned with the radial distance **r** from the origin to the observation point, as illustrated in Figure 2.4. The relationship between **R** and **r** is given by [1]:

$$\mathbf{R} = \left[\mathbf{r}^2 + (\mathbf{r}')^2 - 2\mathbf{r}\mathbf{r}' \cos(\psi)\right]^{1/2} \tag{4.6}$$

An approximation of **R** is provided by [1]:

$$\mathbf{R} = \begin{cases} r - r' \cos\psi & \text{for phase variations} \\ r & \text{for amplitude variations} \end{cases}$$

The far-field region begins approximately at a distance **r** from the antenna, with the following condition for the distance:

$$r \geq \frac{2D^2}{\lambda} \tag{4.7}$$

where:
λ - The wavelength of the antenna, which is approximately the sinusoidal signal of the antenna.
D - The maximum linear dimension of the antenna in [m], where for a bowtie antenna, $D \approx \frac{\lambda}{2}$. Using the **R** in equation 4.8 below and making use of the same pseudocode in

figure 3.11 can get the implementation of the plots of far-field.

$$\lim_{R\to\infty} \tilde{H}(\mathbf{r}) = \frac{e^{-j\beta_0 R}}{R}\tilde{H}_{\text{far}} \tag{4.8}$$

Figure 4.6 shows the results of the scattered magnetic field, **H**-field of the bowtie antenna as a function of θ ranging from $[0 - 360]$.

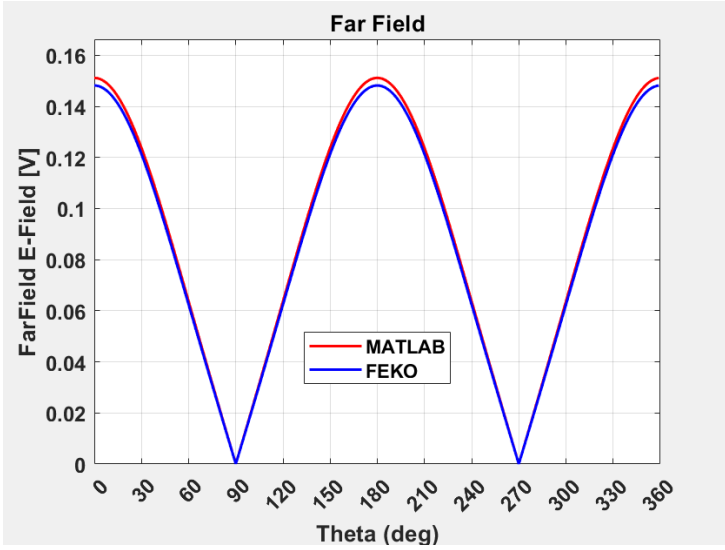

Figure 4.6: Far Field plot for both MATLAB and FEKO for the MoM part

Figure 4.6 shows the comparison of the data obtained from both FEKO and MATLAB, providing insight into the behavior of the far-field of the bowtie antenna showing a strong alignment between the two sets of results validating the accuracy of the integral methods implemented in my MATLAB code and the set-up of my bowtie antenna in FEKO.

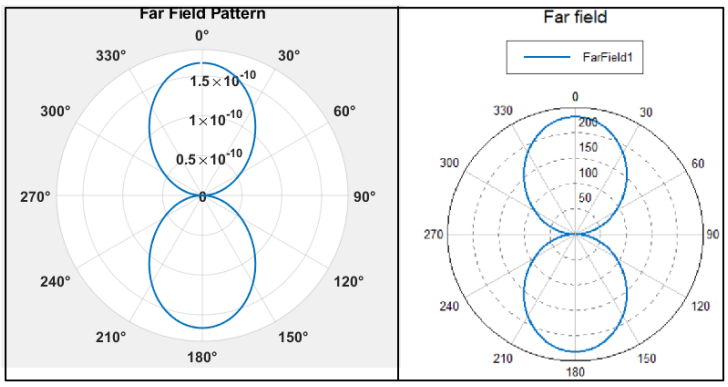

Figure 4.7: Far Field plot for both MATLAB and FEKO in Polar plot

Figure 4.7 compares the far-field radiation patterns obtained using MATLAB (left) and FEKO (right). Both plots show the radiation pattern as a function of angle θ in the far-field region, with consistent results between the two methods. The main lobes are clearly visible around 0° and 180°, indicating a bidirectional radiation pattern typical

for the type of antenna being analyzed. The close agreement between the results from MATLAB and FEKO demonstrates that the analytical approach implemented in the MATLAB code accurately matches the simulation output from FEKO, a widely-used electromagnetic simulation tool. This alignment confirms that the MATLAB code correctly models the far-field characteristics, such as the decay of field strength and the directional behavior, as defined by the far-field conditions.

The consistency between these results suggests that the MATLAB calculations are suitable for practical analysis, while FEKO's detailed simulations provide a robust reference point for validating analytical models.

4.4. Conclusion

This chapter detailed the development and implementation of the MoM solver using a bowtie antenna to compute the radiated magnetic field $H_s^{\mathrm{MoM}}(r)$. Modifications to the original code incorporated field calculations, forming the basis for the hybrid MoM-PO method in Chapter 5.

The process involved initializing the impedance matrix [Z], solving for the current density J^{MoM}, and computing the magnetic field through integration with Green's function. Results showed that the MATLAB code aligned closely with FEKO outputs in both near-field and far-field analyses, confirming its reliability and accuracy for electromagnetic modeling.

Chapter 5

Hybrid MoM-PO solver

5.1. Hybrid MoM-PO solver

In the Hybrid MoM-PO, the MoM bowtie antenna is used as an illuminator for the reflector. My focus is soley on forward coupling, which means that the reflector must be illuminated by the current density $\mathbf{J}_{\text{MoM-PO}}$ as it relates to the Physical Optics (PO) method, without considering the reverse path.

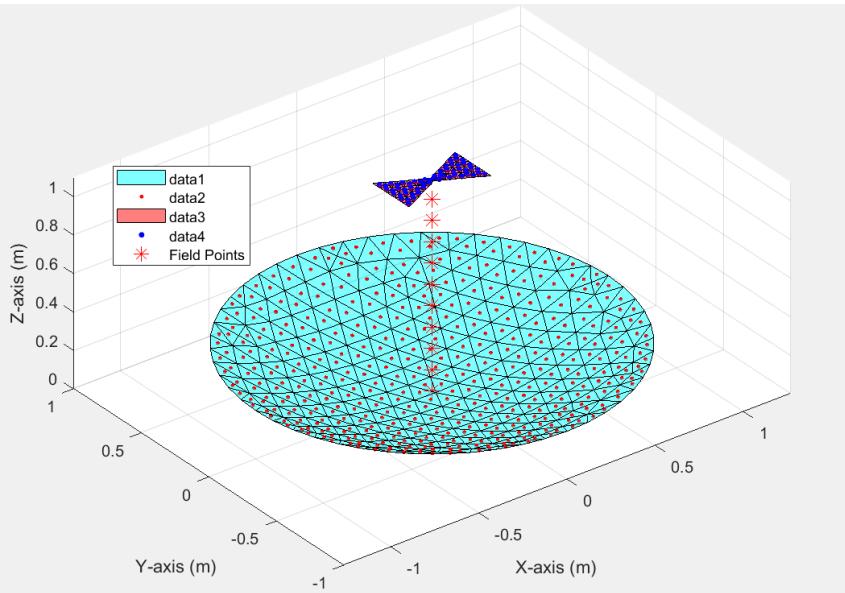

Figure 5.1: Mesh visualization of Hybrid MoM-PO

Figure 5.1 shows the mesh visualization of the Hybrid MoM-PO in MATLAB with the grid of points. In this section discretization of the current is first done for both the PO mesh and the MoM mesh to calculate the induced current density $\mathbf{J}_{\text{MoM-PO}}$ at the centroid of each triangular mesh.

Figure 5.2 presents the pseudo-code for implementing the Hybrid MoM-PO in MATLAB. The key step is calculating the currents on the bowtie antenna using the standard MoM approach.

(a) **Import the Mesh:** Load the PO mesh and the MoM mesh as geometry from a NASTRAN file.

(b) Input the θ and ϕ values for the plane wave excitation.

(c) Write the coordinates of the MoM mesh on the x,y,z axis (location).

(d) **Calculate Centroids and Normals:** Compute the centroid and normal vector $\hat{\mathbf{n}}$ for each triangle.

(e) **Initialize the Impedance Matrix:** Set up the impedance matrix $\mathbf{Z}$ using the provided MoM library.

(f) **Initialize the Voltage Source:** Define the voltage gap source $[\mathbf{V}]$ (edge port in FEKO) for excitation.

(g) Use the calculated Scattered magnetic $\mathbf{H_s^{MoM}}(\mathbf{r})$ field as the incident to the PO mesh:

$$\mathbf{H}^i_{MoM}(\mathbf{r}) = \mathbf{H}^{MoM}_s(\mathbf{r}) \tag{5.1}$$

(h) **Compute the Current Density:** Solve for the current density $\mathbf{J}_{\text{MoM-PO}}$.

$$\mathbf{J}_{\text{MoM-PO}} = 2(\hat{\mathbf{n}} \times \mathbf{H}^i_{MoM}) \tag{5.2}$$

(i) Calculate the gradient of Green's Function using the equation below with $\mathbf{R} = |\mathbf{r} - \mathbf{r}'|$:

$$\nabla G_0(\mathbf{r}, \mathbf{r}') = \frac{(\mathbf{r} - \mathbf{r}')\,(1 + jk_0\mathbf{R})}{4\pi\mathbf{R}^3} e^{-jk_0\mathbf{R}} \tag{5.3}$$

(j) **Scattered Magnetic Field:** Compute the scattered magnetic field $\mathbf{H}_s(\mathbf{r})$ using Green's function:

$$\mathbf{H_s^{MoM}}(\mathbf{r}) = \int\int_S \mathbf{J}_{\text{MoM-PO}}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}') dS'. \tag{5.4}$$

$$\mathbf{H_s^{MoM}}(\mathbf{r}) = \sum_i \mathbf{J}_{\text{MoM-PO}}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r_i'}) \times Area_i \tag{5.5}$$

(k) Plot the near-field and far-field results of the Hybrid MoM-PO comparing MATLAB and FEKO outcomes.

Figure 5.2: Pseudo code for the basic Hybrid MoM-PO implementation in `HybridMoMPOsolver.m`.

The next key step is to cycle through all PO current elements on the reflector and determine the magnetic field $\mathbf{H}^i_{MoM}$ produced by the MoM currents at the center of each element. Use the PO approximation to find the current density on each PO element, which directly corresponds to the current coefficient for that element [2].

This process provides the current distribution on both the bowtie antenna and the reflector. The currents are calculated to generate the radiated field 5.7, resulting in the complete solution for the reflector antenna using the Hybrid MoM-PO approach.

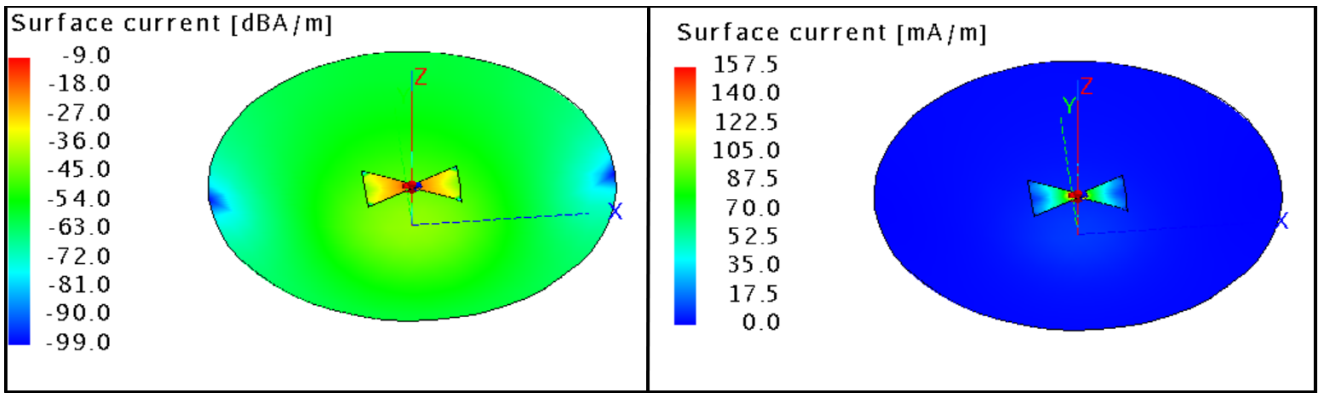

Figure 5.3: Image of the surface current density of the Hybrid MoM-PO

Figure 5.3 presents two images of the surface current density obtained from the Hybrid MoM-PO method. The image on the left uses a scale of [dBA/m], while the picture on the right uses a scale of [mA/m]. Additionally, the figure illustrates how the bowtie antenna illuminates the reflector.

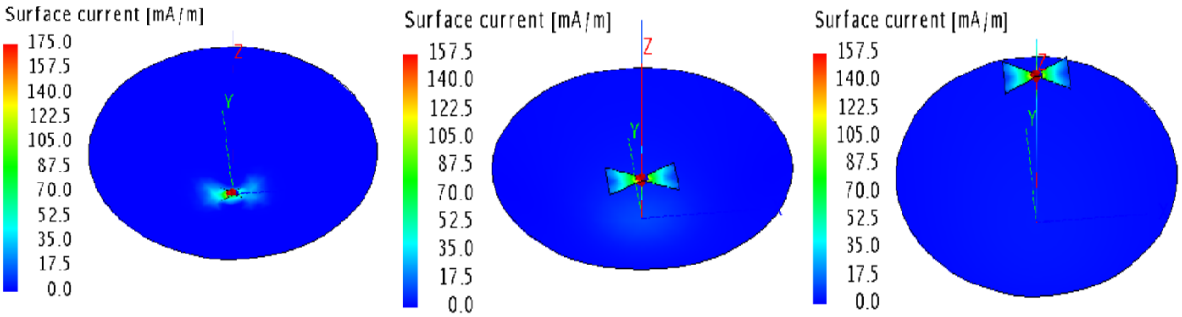

Figure 5.4: 10 observation points

The figure 5.4 above shows how the bowtie antenna illuminates the reflector, with the antenna positioned at different points along the z-axis. The scattered magnetic field $\mathbf{H}_s$ resulting from this interaction will be illustrated through line graphs presented in the near-field and far-field sections.

It is important to note that no shadowing index is required since the entire reflector is assumed to be uniformly illuminated. Additionally, due to time constraints, the shadowing

effect was not implemented in this project. However, if time permits, the figure **??** below shows the ship model from FEKO, where certain areas indicate shadowed regions when illuminated by the bowtie antenna.

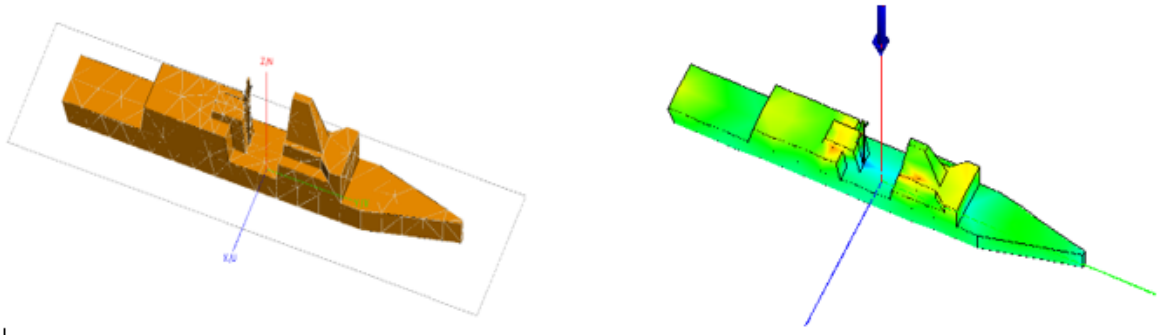

Figure 5.5: Image of the ship with the current and Mesh

The FEKO results demonstrate accurate surface currents, while the results from my MATLAB code may be less precise due to the exclusion of the shadowing effect.

5.2. Near-Field Results

This section presents the near-field results at different observation points with the bowtie antenna carrying along the z-axis. In figure 5.6, the results of the radiated magnetic field $\mathbf{H}_s(\mathbf{r})$ calculated using the approach in 5.7 shows the near-field comparison of both MATLAB and FEKO.

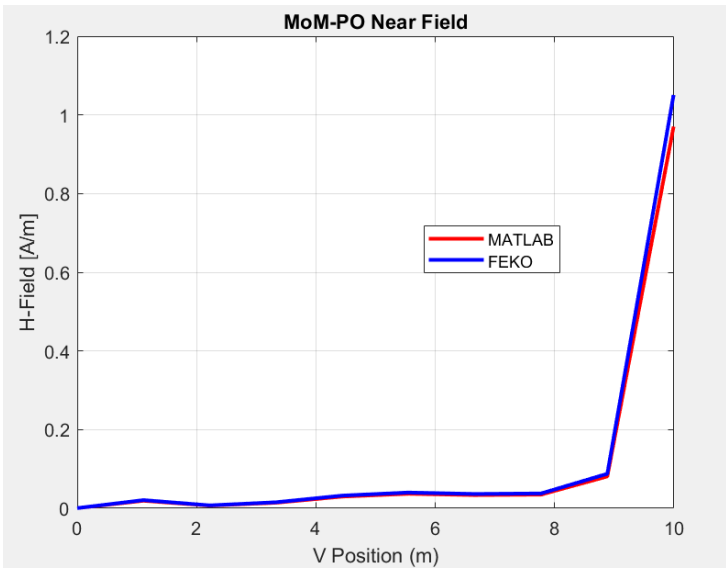

Figure 5.6: Far Field plot for both MATLAB and FEKO

Figure 5.6 shows that the near-field results from the FEKO simulation align well with those produced by my MATLAB code. This close match conforms to the precision of the

integral methods applied in MATLAB for near-field calculations. The agreement between the MATLAB and FEKO results demonstrates that the computational method used in MATLAB, as described in pseudocode 3.8 in Chapter 3 still gives perfect results for the calculation of the near-field using field points.

5.3. Far-Field Results

This section will present the far-field results of both MATLAB and FEKO. For this section, to calculate the far-field you had to follow this steps below:

(a) Calculate the surface current density for both PO and MoM $\mathbf{J}_{\mathrm{PO}}$ and $\mathbf{J}_{\mathrm{MoM}}$

(b) Use the currents to calculate the far-field scattered magnetic field for both PO and MoM:

$$\mathbf{H}_{\mathbf{s}}^{\mathbf{PO}}(\mathbf{r}) = \sum_i \mathbf{J}_{PO}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}_{\mathbf{i}}') \times Area_i \tag{5.6}$$

$$\mathbf{H}_{\mathbf{s}}^{\mathbf{MoM}}(\mathbf{r}) = \sum_i \mathbf{J}_{MoM}(\mathbf{r}') \times \nabla G_0(\mathbf{r}, \mathbf{r}_{\mathbf{i}}') \times Area_i \tag{5.7}$$

(c) Calculate the total far-field scattered magnetic field:

$$\mathbf{H}_s^{\mathrm{FF}}(\mathbf{r}) = \sum \mathbf{H}_s^{PO}(\mathbf{r}) + \mathbf{H}_s^{MoM}(\mathbf{r}) \tag{5.8}$$

(d) Plot the far-field results of the Hybrid MoM-PO comparing MATLAB and FEKO outcomes.

Figure 5.7: Pseudo code for the far-field of the Hybrid MoM-PO

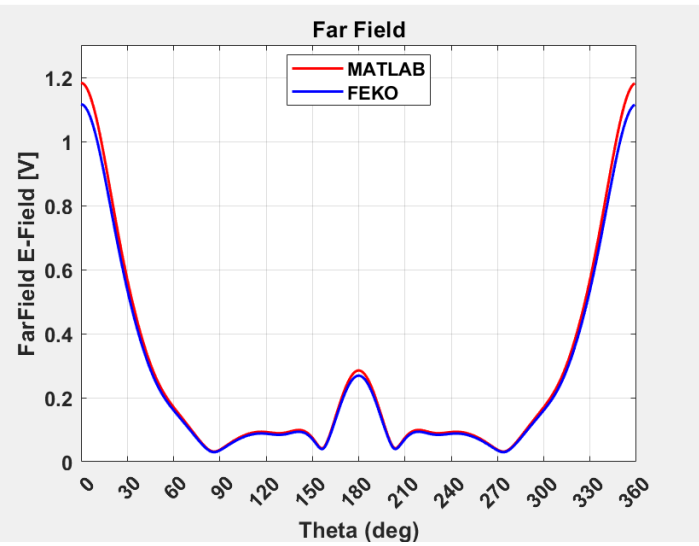

Figure 5.8: Far Field plot for both MATLAB and FEKO

Figure 5.8 above displays the results of the far-field graphs from both MATLAB and

FEKO in a single figure. The alignment of both graphs indicates that the numerical approach and the integral calculation yield accurate results.

5.4. Conlcusion

In this chapter we discussed the Hybrid MoM-PO solver, which models how a bowtie antenna illuminates a reflector using forward coupling. The chapter explained the steps for discretizing the meshes and computing the induced current densities at the triangle centroids.

MATLAB implementation matched FEKO results for near-field and far-field calculations, confirming accuracy. Despite not including shadowing effects, the solver performed well in analyzing electromagnetic interactions. The Hybrid MoM-PO method effectively combines the precision of MoM with the efficiency of PO, proving useful for complex antenna analysis.

Chapter 6

Summary and Conclusion

This report presents a comprehensive examination of various electromagnetic analysis methods, specifically focusing on Physical Optics (PO), Method of Moments (MoM), and their hybridization. It emphasizes the strengths and weaknesses of these approaches, demonstrating their effectiveness in solving complex problems related to electromagnetic scattering, antenna radiation, and interactions with large structures. Foundational concepts such as near-field and far-field analysis are explored.

The development of a PO solver is detailed, highlighting the necessary theoretical and practical steps for calculating scattered magnetic fields using Green's function. Accurate centroid and normal vector calculations for triangular mesh elements are emphasized as crucial for ensuring precision in electromagnetic analysis. The solver's performance is validated through comparisons with FEKO results, showing strong agreement in both near-field and far-field scenarios and affirming the effectiveness of MATLAB in electromagnetic modeling.

The report further discusses the MoM solver, specifically using a bowtie antenna to compute the radiated magnetic field. Modifications to enhance field calculations set the groundwork for the Hybrid MoM-PO approach, which is also introduced. This method effectively models the interaction between the bowtie antenna and a reflector, demonstrating a successful correlation with FEKO results. Overall, the report illustrates how these methodologies can be integrated to enhance the accuracy and efficiency of electromagnetic simulations, paving the way for future advancements in antenna design and analysis.

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Appendix A

Project Planning Schedule

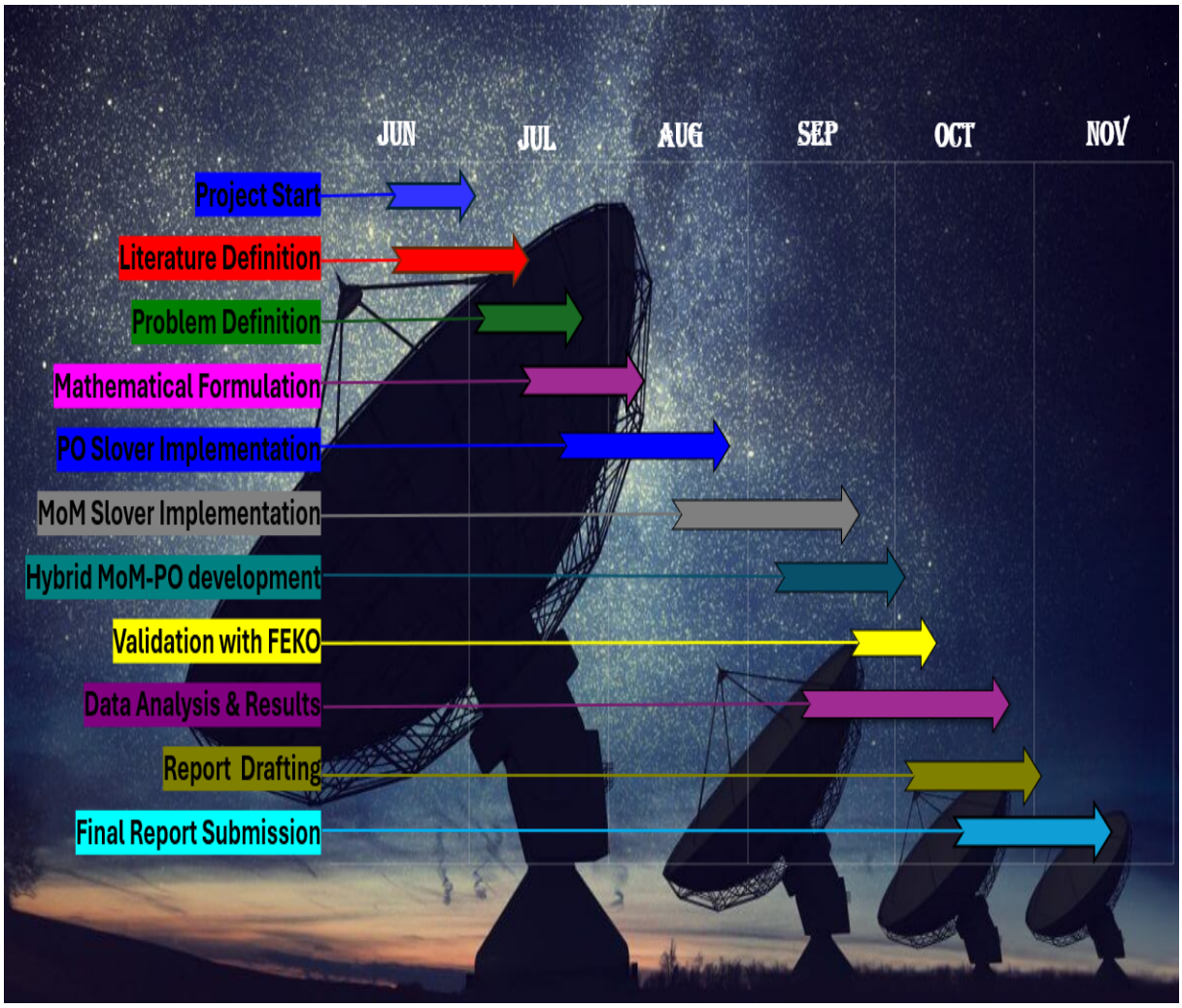

Figure A.1: Figure showing a Gant chart of the planned project schedule

Appendix B

Outcomes Compliance

Table B.1: ECSA Exit Level Outcomes (ELOs) and their achievement by chapter

ELO Description	**How it was achieved**	**Chapter**
GA 1 Problem Solving	This will be accomplished by developing a mathematical model for the scattering problem and implementing it using MATLAB and FEKO. The accuracy of the model will be validated by comparing results from both tools.	3,4,5
GA 2 Application of Scientific and Engineering Knowledge	The project will leverage principles from electromagnetics, applied mathematics, and electrical engineering to calculate the surface current density and scattered field. It will also draw upon knowledge from related fields like signal processing, wave propagation, and electromagnetic compatibility.	2,3
GA 3 Engineering Design	The MATLAB model, incorporating Physical Optics and the Method of Moments, will be developed and tested. The results will then be compared with FEKO simulations to ensure accuracy and efficiency.	3,4,5
GA 4 Investigations, Experiments and Data Analysis	Simulations using MATLAB and FEKO will be conducted to analyze the scattered magnetic field. Various scenarios will be tested to gather data on the field's behavior under different conditions. The collected data will be compared against theoretical predictions and experimental results.	3,4,5
GA 5 Engineering Methods, Skills and Tools, including IT	The project will require proficiency in using engineering tools like MATLAB and FEKO for modeling and simulations. The student will demonstrate competence in these tools throughout the project.	3,4,5
GA 6 Professional and Technical Communication	The project will involve a written report and an oral presentation. These will demonstrate the student's ability to communicate effectively, both orally and in writing.	All
GA 7 Individual Work	The student will take primary responsibility for all aspects of the project, from planning and implementation to analysis and reporting.	All
GA 8 Independent Learning Ability	The student will be required to acquire necessary knowledge independently, whether from literature or online resources, without explicit guidance from the project definition.	All

Appendix C

Mathematical and Physical Constants in Electromagnetics

Parameter	Expression	Name
μ_0	$4\pi \times 10^{-7}\,\mathrm{H/m}$	Permeability of free space
ε_0	$8.85418781761 \times 10^{-12}\,\mathrm{F/m}$	Permittivity of free space
η	$\sqrt{\frac{\mu_0}{\varepsilon_0}}$	Intrinsic impedance of free space
E_0	$1\,\mathrm{V/m}$	Electric field magnitude
H_0	$\frac{E_0}{\eta}$	Magnetic field magnitude
c_0	$\frac{1}{\sqrt{\varepsilon_0\mu_0}}$	Speed of light in a vacuum
f	$300 \times 10^6\,\mathrm{Hz}$	Frequency
λ	$\frac{c_0}{f}$	Wavelength
ω	$2\pi \times f$	Angular frequency
k_0	$\frac{\omega}{c_0}$	Wave number

Table C.1: List of electromagnetic constants, expressions, and their names

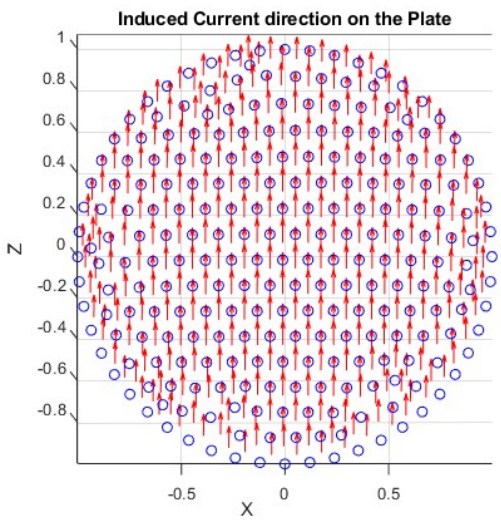

Figure C.1: Induced current direction of the PO

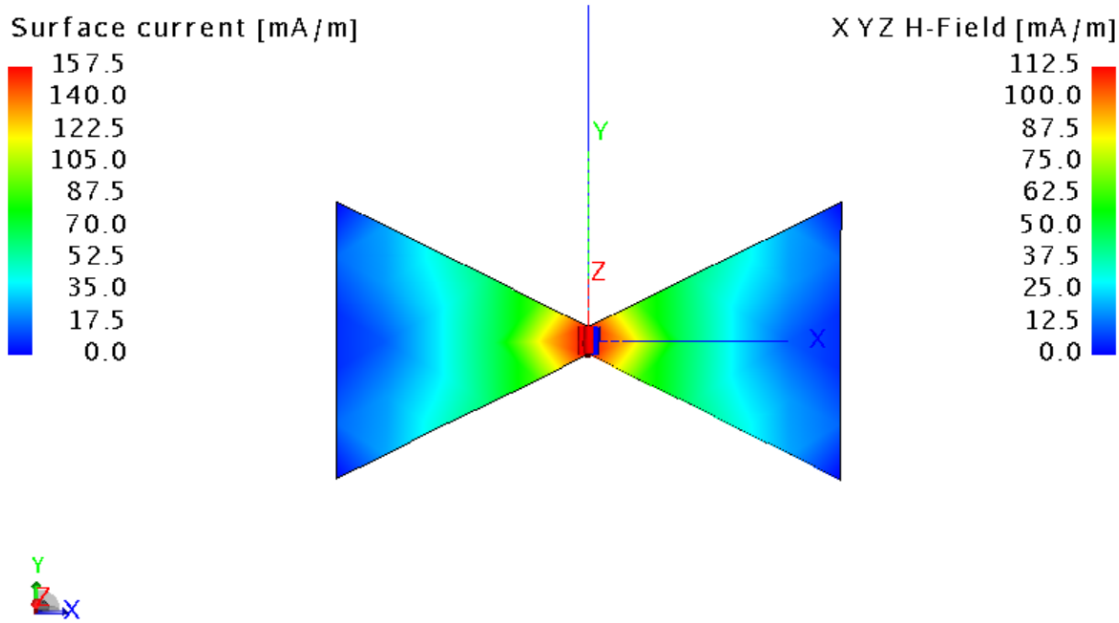

Figure C.2: MoM Bowtie antenna showing $\mathbf{J}_{\mathrm{MoM}}$ and $\mathbf{H}_s^{MoM}$

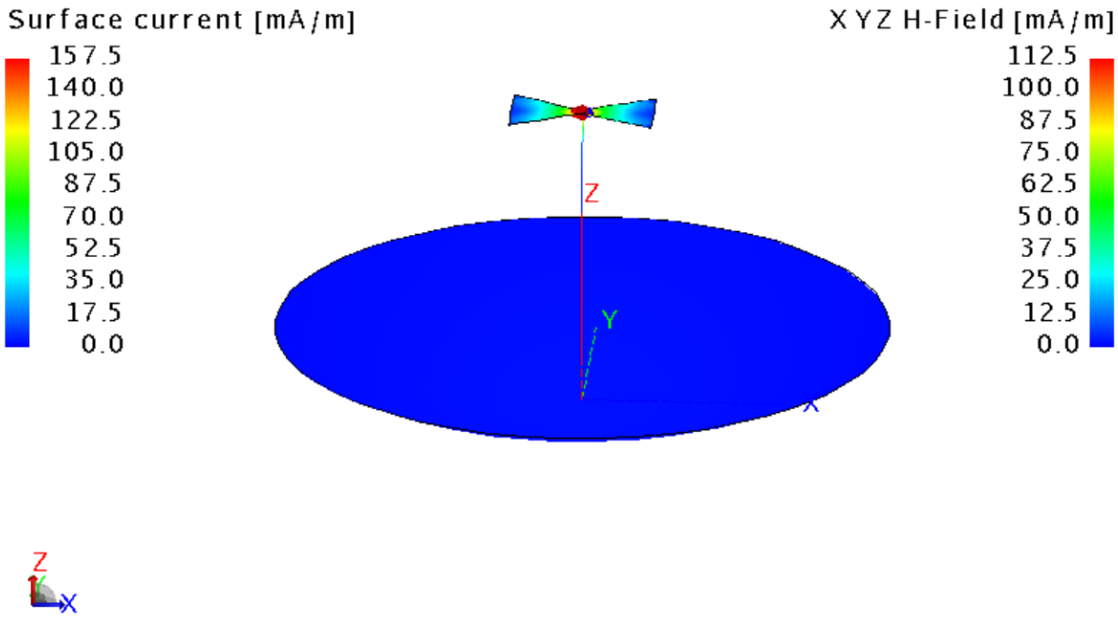

Figure C.3: Hybrid MoM-PO showing $\mathbf{J}_{\mathrm{MoM}}$ and $\mathbf{H}_s^{MoM}$